Quantitative Aptitude

TCS Previous Year Papers and study materials

QUESTIONS

Q 1.2/3rd of the balls in a bag are blue, the rest are pink. if 5/9th of the blue balls and 7/8th of the pink balls are defective, find the total number of balls in the bag given that the number of non defective balls is 146?

a) 216

b) 649

c) 432

d) 578

Solution- let total no of balls =x

blue=2x/3

pink=x/3

total no of defective balls = 10x/27 +7x/24

=143x/216

non defective balls=x-143x/216=146

x=432

Q 2. Find no of ways in which 4 particular persons a,b,c,d and 6 more persons can stand in a queue so that A always stand before B. B always stand before C, And C always stand before D.

a)6!

b)7!

c)1006*6!

d)10046!

Solution - a,b,c,d are grouped ie consider them as one and remaining as 6.

total 6+1 = 7! Ways

Q 3. 100 students appeared for two different examinations 60 passed the first,50 the second and 30 both the examinations. Find the probability that a student selected at random failed in both the examination

a)5/6

b)1/5

c)1/7

d)5/7

Solution-60+50-30=80

100-80=20

20/100=1/5.

so B is the answer.

Q 4. There are 10 points on a straight line AB and 8 on another straight line AC none of them being point A. how many triangles can be formed with these points as vertices?

Option

a. 680

b. 720

c. 816

d. 640

Solution-

To form a triangle we need 3 points

select 2 points from the 10 points of line AB & 1 from the 8 on AC

= (10C2)*(8C1)

select 2 points from the 8 points of line AC & 1 from the 10 on AB=

(8C2)*(10C1)

total no. of triangles = (10C2)*(8C1)+ (8C2)*(10C1) = 640

d.640

Q 5. From a bag containing 8 green and 5 red balls,three are drawn one after the other .the probability of all three balls beings green if the balls drawn are replace before the next ball pick and the balls drawn are not replaced , are respectively.

a)512/2197,336/2197

b) 512/2197,336/1716

c) 336/2197,512/2197

d) 336/1716,512/1716

Solution-THE PROBABILITIES OF GETTING WITH REPLACEMENT IS=8/13*8/13*8/13=512/2197

THE PROBABILITIES OF GETTING WITHOUT REPLACEMENT =8/13*7/12*6/11=336/2197

Q 6. find the greatest number that will divide 148 246 and 623 leaving remainders 4 6 and 11 respectively.

a) 20 b) 12 c)6 d)48

Solution-Hcf ((148-4), (246-6), (623-11))=12

Q 7. a mother her little daughter and her just born infant boy together stood on a weighing machine which shows 74kgs.how much does the daughter weigh if the

mother weighs 46kg more than the combined weight of daughter and the infant and the infant weighs 60% less than the daughter.

a)9

b)11

c)cannot be determined

d)10

Solution-daughter weight is x

infant weight is 60% less than daughter i.e., 0.4x

mother weight is (x+0.4x+46)

total weight = x + 0.4x + (x+0.4X+46) = 74

solving the eq. x = 10

option d is correct

Q 8. find the number of ways a batsman can score a double century only in terms of 4's & 6's?

a)15

b)16

c)17

d)18

Solution-4's and 6's

50 0

47 2

44 4

41 6

38 8

35 10

32 12

29 14

26 16

23 18

20 20

17 22

14 24

11 26

8 28

5 30

2 32

So total 17 ways but here it is 4's & 6's both so don't consider 1st one

Final ans : 16 ways

Q 9. 98. Thomas takes 7 days to paint a house completely whereas Raj would require 9 days to paint the same house completely. How many days will take to paint the house if both them work together. (Give answers to the nearest integer)?

A. 4

B. 2

C. 5

D. 3

Solution-work done by thomas in day=1/7

work done by other in a day=1/9

work done by both in a day=1/7 +1/9 =16/63

days required by both = 63/16

Answer-A

Q 10. how many positive integers less than 4300 of digits 0-4.

a) 560 b)565 c)575 d)625

Solution-Ans is 575

one digit no =4 (0 is not a positive integer)

two digit no=4*5=20

three digit no=4*5*5=100

four digit no=3*5*5*5=375(the possibility for 1,2,3 will come in the first position)four digit no=1*3*5*5(the possibility of 4is fixed in the first position and then 0,1,2 is comes in second position)and the last digit is 4300 we include this number also

Ans is 4+20+100+375+75+1=575

Q 11. A person travels from Chennai to Pondicherry in cycle at 7.5 Kmph. Another person travels the same distance in train at a speed of 30 Kmph and reached 30 mins earlier. Find the distance.

A)5 Km

B)10 Km

C)15km

D)20km

Solution-Let, time taken by 🚲 =t

//ly, time taken by train 🚆 =t-30

We know that.....speed=distance /time

Speed of bicycle, 7.5=d/t. ..(1)

Speed of , 30=d/(t-30/60)....(2)

Sol 1&2,we get t=0.666

By sub and value in equal(1)

We d=4.999~5km

Q 12. A bag contains 8 white balls, and 3 blue balls. Another bag contains 7 white, and 4 blue balls. What is the probability of getting blue ball?

a)3/7 b)7/22 c)7/25 d)7/15

Solution-First we have to select a bag and then we will draw a ball.

Probability of selection of both bags is equal =1/2

Now probability of blue ball taken from first bag = (1/2) x (3/11)

and probability of blue ball taken from second bag = (1/2) x (4/11)

So probability of blue ball = (1/2) x (3/11) + (1/2) x (4/11) = 7/22

Q 13. In a 3*3 square grid comprising 9 tiles each tile can be painted in red or blue color. When the tile is rotated by 180 degree, there is no difference which can be spotted. How many such possibilities are there?

a) 16 b)32 c)64 d)256

Solution-ans is 32 .

ans grid has to be rotated at 180 degrees.

hence,

11=33

12=32

13=31

21=23

22=22

31=13

32=12

33=11

cell 11-33 can be altered in 2 ways (as thr are 2 colours)

cell 12-32 can be altered in 2 ways

cell 13-31 can be altered in 2 ways

cell 21-23 can be altered in 2 ways

and , cell 22 can be altered in 2 ways

, so

2X2X2X2X2 = 32.

Q 14. Jake can dig a well in 16 days. Paul can dig the same well in 24 days. Jake, Paul and Hari together dig the well in 8 days. Hari alone can dig the well in

option

A)48

B)96

C)24

D)32

Solution-1/x = 1/8 - (1/16 + 1/24)

so x=48

ans = 48 days

Q 15. For any two numbers we define an operation $ yielding another number, X $ Y such that following condition holds:

- **X $ X = 0 for all X**
- **X $ (Y $ Z) = X $ Y + Z**

Find the Value of 2012 $ 0 + 2012 $ 1912

Options

a) 2112

b) 100

c) 5936

d) Can not be determined

Solution-here $ represent the - operator so X-X=0 first condition

x$(y$z)=x-(y-z)=x-y+z

it can be written as x$y+z

2012-0+2012-1912=2112 so ans is 2112

Q 16. On a toss of two dice, A throws a total of 5. Then the probability that he will throw another 5 before he throws 7 is

a)40% b) 45% c)50% d)60%

Solution-ans: 40%

Explanation:

total probabilities for getting 5 = 4/36

total probabilities for getting 7 = 6/36

Total Probability = 10/36

We need only 5, hence prob of getting only 5 is (4/36)/(10/36)

=40%

Q 17.

1,2,2,3,3,3,4,4,4,4,1,1,2,2,2,2,3,3,3,3,3,3,4,4,4,4,4,4,4,4,……..

Then what is the 2320 position of the number in the sequence?

a) 2 b) 1c) 3 d) 4

Solution-answer is b)1

1,2,3,4(1-1time 2- 2times 3-3 4-4)=10 terms /completes cycle and starts from 1

1,2,3,4(1-2 2-4 3-6 4-8)= 20 terms /completes cycle and starts from 1

(1,2,3,4)each digit 3 time to its value =30 terms/completes cycle and starts from 1

10+20+30+40+50+...... **=x**

x is nearer value to 2320 solving n(n+1)/2

10.(21.22)/2= 2310

analysing it 2310 completes cycle and starts from 1 again

now it 22 times

(1-22 times 2-44 times.....**)**

2320 position will occupied by 1

Q 18. In 2003, there are 28 days in February and there are 365 days in the year. In 2004, there are 29 days in February and there are 366 days in the year. If the date

March 11, 2003 is a Tuesday, then which one of the following would be the date March 11, 2004 be?

A. Wednesday

B. Tuesday

C. Thursday

D. Monday

Solution-Every year day is increased by 1 odd day. Or in leap year it is increased by 2 odd days.

so 11 march 2003 is tuesday,

11 march 2004 is thursday

C. Thursday

Q 19. How many 6 digit even numbers can be formed from digits 1 2 3 4 5 6 7 so that the digit should not repeat and the second last digit is even?

a)6480 b)320 c)2160 d)720

solution-given 6th digit even number , so last digit 2 or 4 or 6-> 3 ways

" 5th digit should be even...so there will be 2 ways(rep. not allowed)

so,therefore we get 5*4*3*2*2*3=720 ways

Q 20. There are 5 letters and 5 addressed envelopes. If the letters are put at random in the envelops, the probability that all the letters may be placed in wrongly addressed envelopes is.

a)119 b)44 c)53 d)40

Solution-If there is one letter and one envelope then no way you can put it wrong(S1).

If there are 2 letters and 2 envelopes then you can put them wrong in 1 way(S2).

If there are 3 letters and 3 envelopes then you can put them wrong in 2 ways(S3).

If there are 4 then you can put them wrong in 9 ways(S4).

If there are 5 then you can put them wrong in 44 ways(S5).

If you observe you can find a pattern.

S3=(S1+S2)*2

S4=(S2+S3)*3

S5=(S3+S4)*4

S6=(S4+S5)*5

In general, Sn=(Sn-2 + Sn-1)*(n-1)

So, if there are 5 letters then S5=(S3+S4)*4=(2+9)*4=44

Q 21. How many liters of a 90% of concentrated acid needs to be mixed with a 75% solution of concentrated acid to get a 30 liter solution of 78% concentrated acid?

a) 8 b)9 c) 7 d)6

Solution-the concentration is given which is wrt 100, hence we can take x lt of 90% and (30-x)of 75%

x*90 + (30-x)*75 = 30*78 hence the ans is 6 ltr...

Q 22. Average marks of a,b,c is 48. When d joins average becomes 47. E has 3 more marks than d. Average marks of b,c,d,e is 48. What is the marks of a?

a) 42 b) 43 c) 53 d)56

Solution- a+b+c=144

a+b+c+d=188

d=44

e=47

b+c+d+e=192

b+c=101

b=43 ans.

Q 23. On a certain assembly line, the rejection rate for Hyundai i10s production was 4 percent, for Hyundai i20s production 8 percent and for the 2 cars combined 7 percent. What was the ratio of Hyundais i10 production?

option

a) 3/1

b) 2/1

c) 1/1

d) ½

Solution-let the no of i10 cars be x and i20 be y

now th rejcted i10 cars are 4x/100 and i20 cars are 8y/100

and it is given that 4x/100 +8y/100 = 7(x+y)/100

so we get y = 3x

so the ratio is 3:1

Q 24. For a car there are 5 tyres including one spare tyre(4+1). All tyres are equally used. If the total distance travelled by the car is 40000km then what is the average distance travelled by the each tyre?

Option

a) 10000

b) 40000

c) 32000

d) 8000

Solution-total distance travelled by the car=40000km

so total distance travelled by 4 wheels=4*40000=160000

as all tyres(4+1) are equally used

so average distance travelled by the each tyre=160000/5=32000

option(c)

Q 25. If A=x3 y2 and B=xy3, then find the HCF of A,B.

a) x4y5 b)xy2 c)xy d)x3

Solution-if A=x^3*y^3

B=xy3

then hcf=xy2

bcoz hcf is always equal to the highest common powers between the expressions,

i.e x1 is common in both and also y2 is common in both.

Q 26. In a clock the long hand is of 8cm and the short hand is of 7cm. if the clock runs for 4 days find out the total distance covered by both the hands

a) 1824 π cm b)1648π cm c)1724π cm d)2028π cm

Solution-Explanation: Short Hand (Hour hand)

Hour hand makes a full rotation in 12 hours.

One full rotation in 12 hours =>2πr=14π cm traversed every 12 hours.

For one day(24 hours), we have 28π cm, twice that of a 12 hour period.

For 3 days, we then have 4 x 28 π=112π cm traversed.

Long Hand (Minute hand)

One full rotation in 1 hour =>2πr=16π cm traversed every hour.

For one day, we have 24 x 16 π=384π cm.

For 4 days, we then have 4 x 384π=1536π cm traversed.

Total Distance

For the total, we have 112π+1536π=1648π cm.

Q 27. A can finish a work in 18 days and B can do the same work in 15 days. B worked for 10 days and left the job. In how many days A alone can finish the remaining work?

a) 7 b) 6 c) 5 d) 10

Solution-A can finish 118118th of the total work in a day and B can finish 115115th of the total work in a day. After working 10 days by B alone,the work remains is 1−1015=131−1015=13th of the total work.

It can be finished by A in13÷118=1×183×1=613÷118=1×183×1=6days.

Q 28. In how many way possible ways can you write 1800 as a product of 3 positive integers a,b,c

a)350 b)360 c)380 d)450

Solution-1800=2^3*3^2*5^2

then 5c2*4c2*4c2=360 ans

Q 29. In ahorse racing competition there were 18 numbered 1 to 18.The organizers assigned a probability of winning the race to each horse baesd on horses health and training the probability that horse one would win is 1/7, that 2 would win is 1/8, and that 3 would win is 1/7.Assuming that tie is imposible Find the chance that one of these three will win the race?

a)22/392 b)1/392 c)23/56 d)391/392

Solution-HORSE 1: 1/7 WINNING PROBABILITY

HORSE 2: 1/8 WINNING PROBABILITY

HORSE 3: 1/7 WINNING PROBABILITY

ONE OF THESE WIN THE RACE:

=> 1/7 + 1/8 + 1/7

=> 8/56 +7/56 + 8/56 (TAKING LCM)

=> 23/56

Q 30. Apple costs L rupees per kilogram for first 30kgs and Q rupees per kilogram for each additional kilogram. If the price of 33 kilograms is 11.67and for 36kgs of Apples is 12.48 then the cost of first 10 kegs of Apples is

a) 3.50 b) 10.53 c) 1.17 d)2.8

Solution-30L+3Q=11.67

30L+6Q=12.48

3Q=.81 Q= .27

from that L=0.362 cost of 10 kg apple is 10*.362=3.6 a) 3.50

Q 31. How many vehicle registration plate numbers can be formed with digits 1,2,3,4,5(no digits being repeated)if it is given that registration number can have 1 to 5 digits ?

a) 205 b) 100 c) 325 d) 105

Solution-you can have registration plates of 5,4,3,2 or 1 digits

So, it's 5*4*3*2*1 + 5*4*3*2 + 5*4*3 +5*4 + 5

=120 + 120 + 60 + 20 +5

=> 325

Q 32. Jake and Paul each walk 10 km. The speed of jack is 1.5 faster than paul speed . what is the Jake's speed ?

a)4 b)6 c)7 d)8

Solution-Let pauls speed be x kmph

Then jacks speed is (x+1.5) kmph

(10/x)-(10/1.5+x)=1.5

X=2.5

Jacks speed=2.5+1.5=4kmph

Q33. In this question A^B means A raised to the power B. If f(x)=a*x^4-b*x^2+x+5. f(-3)=2. Then f(3)=?

a)3 b) 7 c) 8 d) 6

Solution-Given that f(x)=a*x^4-b*x^2+x+5

and f(-3)=2

so => a*(-3)^4-b*(-3)^2+(-3)+5=2

=> 81a-9b+2=2

=> 81a-9b=0 ------- equation(1)

now f(3)=a*(3)^4-b*(3)^2+3+5

=81a-9b+8

and from equation (1) 81a-9b=0

so f(3)=0+8=8

Q 34. Of a set of 30 numbers, average of first 10 numbers = average of last 20 numbers. Then the sum of the last 20 numbers is ?

(a) 2 X sum of first ten numbers

(b) 2 X sum of last ten numbers

(c) Sum of first ten numbers

(d) Cannot be determined with given data

Solution-since, average=(sum of n no.s)/(total no)

therefore, (sum of first 10 no.s)/10 ==(sum of last 20 no)/20

hence. (sum of last 20 no.s) = 2*(sum of first 10 no.s)

Q 35. Mother, daughter and infant total weight is 74 kg. Mother's weight is 46 kg more than daughter and infant's weight. Infant's weight is 60% less than daughter's weight. Find daughter's weight.

a)10 b)9 c)8 d) 7

Solution-Total Age is M+D+I=74

given that M-D-I=46

solving above 2 eq's we get Mother age = 60

now remaing age=14 which is sum of daughter and Infant age.

Given that Infant age is 60% lass than Daughter.

i.e If daughter age is 100 then infant age is 40.

So ages ratio Of D and I is 100 : 40 i.e ... 5 : 2

So (5+2)=7 parts equal to 14.

then 5 parts equal to 10.

2 parts equal to 4.

daughter age is =10

and infant age is = 4

Answer a) 10

Q 36. In a certain city, 60% of the registered voters are congress supporters and the rest are BJP supporters. In an assembly election, if 75% of the registered congress supporters and 20% of the registered BJP supporters are expected to vote for candidate A, what percent of the registered voters are expected to vote for candidate A?

a) 53 b) 20 c) 60 d) 75

Solution-let the people in the city be x

60% are congress=60% of x

40% are BJP=40% of x

out of 60%,75% voted for congress=75%(60% of x)=18x/40

out of 40%,20% voted for congress=20%(40% of x)=8x/100

total=18x/40+8x/100=106x/200

total percent=106x/200*100=53% of x

Q 37. In a particular year the month of january had exactly 4 thursdays and 4 sundays , on which day of the weekk, jan 1 occurs?

a) Monday b) Tuesday c) Thursday d) Wednesday

Solution-as there are 4 fulls weeks i.e 28 days..

so..every day occurs min 4 times.

den remaining 3 days (as jan has 31 days) will be monday tuesday wednesday.

so on 31st jan comes wednesday.

so 1st jan ll be MONDAY

Q 38. A number when divided successively by 4 and 5 leaves remainder 1 and 4 respectively. when it is successively divided by 5 and 4, then the respective remainders will be;

a). 1,2

b). 2,3

c). 3,2

). 4,1

Solution-When dividing a positive integer nn by another positive integer DD (divider), we obtain a quotient QQ, which is a non-negative integer and a remainder R, which is an integer such that 0≤R<D0≤R<D. We can write n=DQ+R.n=DQ+R.

When dividing our number nn by 4 we obtain a remainder of 1, so, if the quotient is some integer QQ, we can write n=4Q+1.n=4Q+1.

Now, dividing QQ by 5, we obtain another quotient say qq and remainder 4, thus we can write Q=5q+4.Q=5q+4.

It follows that n=4(5q+4)+1=20q+17.n=4(5q+4)+1=20q+17.

Since n=20q+17=5(4q+3)+2n=20q+17=5(4q+3)+2, it means that when dividing nn by 5 first, we get a quotient 4q+34q+3 and remainder 2.

Then dividing 4q+34q+3 by 4 we obviously obtain a remainder of 3.

Q 39. Average salary of 17 teachers is 45000.3 teachers left and the average salary dropped by 2500.What is the sum of salaries of 3 teachers who left?

a)173000 b) 176000 c) 170000 d) 85000

Solution- Total Initial Salary : 17*45000 = 765000

Average Salary After removal of 3 Teachers = 45000-2500 = 42500

Total Final Salary : 14*42500 = 595000

Sum of Salaries of 3 teachers who left : 765000 - 595000 = 170000

Q 40. There are 20 persons sitting in a circle. In that, there are 18 men and 2 sisters. How many arrangements are possible, in which the two sisters are always separated by a man?

a. 18!*2

b. 17!

c. 17!*2

d. 12

Solution- 18!*2 Consider 1 man along with two sisters as one group.. so they can be arranged in 17! ways as circular.. The one man in between the two sisters can be out of any 18 men.. so,17!*18. and the two sisters can be arranged in 2 ways. so 18!*2

Q 41. a number plate can be formed with two alphabets followed by two digits with no repetition. then how many possible combinations can we get?

a) 58500 b) 67600 c) 57850 d) 58761

Solution- no.of alphabets=26 (a-z), no.of digits=10(0-9).

ways of arranging two alphabets with out repetition=26*25;

ways of forming two digits without repetition=10*9

no.of combinations of forming the number on number plate= 26*25*10*9=58500

Q 42. The letters in the word "PLACES" are permuted in all possible ways and arranged in the alphabetical order.Find the word at 48 position.

a) AESPCL

b)ALCEPS

c)ALSCEP

d)AESPLC

Solution- ans = (d)

for words AC** possible ways for arranging * will be 4!=24**

now next seq in alphabatical order will be AE** so....**

same way for AE** possible ways for arranging * will be 4!=24**

..

..

Thus, 48th element will be last element in AE** that means AE followd by reverse alphabatical order! which is AESPLC**

Q 43. A alone can do 1/4th of the work in 2 days. B alone can do 2/3th of the work in 4 days. If all the three work together, they can complete it in 3 days so what part of the work will be completed by C in 2 days?

A. 1/12

B. 1/8

C. 1/16

D. 1/20

Solution- A can do the total work in 8 days, and B can do it in 6 days.

Let the total work be 24 units. Now capacities are

A = 24/8 = 3,

B = 24/6 = 4,

A + B + C = 24/3 = 8

So Capacity of C = 1 unit.

In two days C will do 2 units which is 2/24th part of the total work. So 1/12th part.

Q 44. A sum is sufficient to pay either George age for 15 days or marks wage to 10dayshow long together?

a) 5 b) 6 c) 8 d) 9

Solution- George one day work is (1/15)

Marks one day work is (1/10)

G+M=(1/15)+(1/10)=(1/6)

6 is the answer

Q 45. In how many ways a team of 11 must be selected a team 5 men and 11 women such that the team must comprise of not more than 3 men.

a) 1565 b) 2256 c) 2456 d) 1243

Solution- Maximum 3 men can be played which means there can be 0, 1, 2, 3 men in the team.

(5C0×11C11)+(5C1×11C10)+(5C2×11C9)+(5C3×11C8)=2256

Q 47. how many prime numbers less that 100 and greater than 3 are of the form:4x+1 , 5y-1

a) 11 b)12 c)7 d)None of the above

Solution- all the prime numbers between 3 and 100 are:5,7,11,13,17,19,23,29,31,37,41,43,47,53,59,61,67,71,73, 79, 83, 89 and 97

numbers must end with 4 or 9 to follow 5y-1 condition.

so the numbers are:29,89

so there are two numbers.

Q 48. Three dice are rolled. What is the probability of getting a sum of 13?

a) 19/216 b) 21/216 c) 17/216 d) 23/216

Solution- Just count the number of ways to get 13. We just need to count possibilities for two dice because the third dice value is fixed. For two dice, the sum can be anywhere from 7 to 12 and that would be 6 + 5 + 4 + 3 + 2 + 1 = 21. So, the probability is 21/216

Q 49. On a 26-question test, five points were deducted for each wrong answer and eight points were added for each correct answer. If all the questions were answered, how many were correct if the score was zero?

a). 10

b). 11

c). 12

d). 13

Solution- if x no. of question is wrong and y no. of question is correct then

-5*x+8*y=0 and x+y=26

by solving we get

x=16 & y=10

Q 50.Two alloys A and B are composed of two basic elements. The ratios of the compositions of the two basic elements in the two alloys are 5:3 and 1:2, respectively. A new alloy X is formed by mixing the two alloys A and B in the ratio 4:3. What is the ratio of the composition of the two basic elements in alloy X?

a)1:1 b) 2:3 c) 5:2 d) 4:32

Solution- Let the actual amount of A be 4x and actual amount of B be 3x since A and B are in the ratio 4/3

amount of first basic element in the new alloy T (5/8)*4x + (1/3)*3x = (7x)/2

amount of second basic element in the new alloy T (3/8)*4x + (2/3)*3x= (7x)/2

so ratio of first basic element to second basic element: [(7x)/2] / [7x/2] = 1/1 = 1:1 (Answer A)

Q 51. Babla alone can do a piece of work in 10 days ashu alone can do it in 15 days. The total wages for the work in Rs.5000 .how much should babla be paid of they work together for entire duration of the work

a)4000 b)3000 c)5000 d)2000

Solution- babla 10 day ------now 30/10=3 unit per day

L.C.M = 30(total work they have to perform)

ashu 15 day -------30/15=2 unit per day

so their ratios of work is 3:2

so babla will get 3/5*5000=3000

Q 52. Average of 3 numbers ABC is given as 48. Average of A,B,C,D is 46. Its given that E is having 3 more than D, then Average of B,C,D,E is 45. What is the score of A?

a) 46 b) 48 c) 49 d) 47

Solution- average A+B+C= 48; A+B+C=3*48=144;

average A+B+C + D= 46; A+B+C+D =4*46=184;

D=40; GIVEN E=D+3=40+3=43;

B+C+D+E=45*4= B+C+40+43=45*4

B+C=180-40-43

B+C=97

A+B+C=144

A+97=144

A=47

Q 53. Raj travels a part of journey by taxi paying 15 per km and rest by train paying 21per km. If he travels a total of 450 Km and pay Rs.8130 then the distance travelled by raj in train?

a) 230 b) 260 c) 190 d) 180

Solution- let the distance traveled by train =x

therefore distance travelled by taxi = (450-x)

now, 15*(450-x) + 21*x=8130

or, x=230

Q54. A sum is sufficient to pay either George age for 15 days or marks wage to 10dayshow long together?

a) 9 b) 5 c) 6 d) 8

Solution- George one day work is (1/15)

Marks one day work is (1/10)

G+M=(1/15)+(1/10)=(1/6)

6 is the answer

Q 55. Radius of the bigger circle is 1. Which area will be greater?

a) 5 b) 4 c) cannot be determined d) none of these

Solution- If the radius of the bigger circle is 1, then diameter = 2 units. Which in turn equals to diagonal of square.

Let the side of the square be x. Then x2 + x2 = 22 ⇒ 2x2 = 4 ⇒ x = √22

Now diameter of the inner circle = side of the square. So radius of the inner circle = √22=1√222=12

Areas marked by 1, 2, 3, 4 = (Area of the circle - area of the square)/4 = π(1)2−(√2)24π(1)2−(2)24 = π−24π−24 = 0.285

Areas marked by 5, 6, 7, 8 = (Area of the square - area of the inner circle)/4 = (√2)2−π(1√2)24(2)2−π(12)24 = 2−π242−π24 = 0.1075

So Area marked by 4 is bigger.

Q 56. University of Vikram has enrolled nine Phd candidates.Babu Chitra,Dheeraj , Eesha,Farooq,Gowri , Hameed,Iqbal,Jacob.

-Farooq and Iqbal were enrolled on the same day as each other, and no one else was enrolled that day.

-Chitra and gowri were enrolled on the same day as each other, and no one else was enrolled that day.

-On each of the other days of hiring , exactly one candidate was enrolled.

-Eesha was enrolled before Babu.

-Hameed was enrolled before Dheeraj

-Dheeraj was enrolled after Iqbal but before Eesha

- Gowri was enrolled after both Jacob and Babu

-Babu was enrolled before Jacob

Who were the last two candidates to be enrolled?

a)Gowri and Chitra b) Babu and Chitra c) Babu and Gowri d) Eesha and Jacob

Solution-

Given that

1. Easha < Babu

2. Hameed < Dheeraj

3. Iqbal < Dheeraj < Easha

4. Jacob/Babu < Gowri

5. Babu < Jacob

from 1 and 5, Easha was before Babu and Jacob so she cannot be in the last two. Option B ruled out

from 4 and 5, babu is before Jacob and Gowri so he cannot be in the last two. Options a, c ruled out.

So option d is correct.

Q 56. A boy buys 18 sharpeners (brown or white) for Rs. 100. For every white sharpener, he pays one rupee more than the brown sharpener. What is the cost of white sharpener & how much did he buy?

a) 5,13 b)5,10 c)6,10 d) None of these

Solution- If he bought x white sharpeners @ Rs (y+1) and (18-x) brown sharpeneres @ Rs y per sharpener, then

x*(y+1)+x*(18-x)=100

x= 100-18y

Only integral value of x less than 18 will be 10.

then x=10, y=5

so he bought 10 white sharpeners @ Rs 6 per sharpeners and 8 brown sharpeners @ Rs 5 per sharpener.

c)6,10

Q 57. If M is 30% of Q, Q is 20% of P, N is 50% of P, Then M/N = ?

a) 4/3 b) 3/25 c) 6/5 d) 3/250

Solution- m=3/10q,q=1/5p,n=1/2p

here we can find d value of p and put into d second to get value of q

so m/n will be 3/25

Q 58. In a staircase, there are 10 steps. A child is attempting to climb the staircase. Each time, she can either make 1 step or 2 steps. In how many different ways can she climb the stair case?

a). 10 b). 21 c). 36 d). None of these.

Solution- she can go by

1 steps-1 way

that is choosing 1 two-step in 9 moves:

9C1 : 9 ways//

2 two-steps:

choosing 2 two-steps in 8 moves:

8C2 = 28 ways//

3 two-steps

7C3 = 35 ways//

4 two-steps//

6C4 = 15 ways//

5 two-steps//

which covers all the 10 stairs.. that means only one way

2 2 2 2 2 = 1 way//

Adding all the ways:

1 + 9 + 28 + 35 + 15 + 1 = 89 ways//

Q 59. Eesha bought two varities of rice costing 50Rs per kg and 60 Rs per kg and mixed them in some ratio.Then she sold that mixture at 70 Rs per kg making a profit of 20 % What was the ratio of the mixture?

)1:10 b) 3:8 c) 1:5 d) 2:7

solution- let x be the price of the rice after mixing,then

120*x/100=70

x=175/3 by eligation method

the required ratio is(60-175/3):(175/3-50)=1:5

Q 60.Find the 32nd wordin the list where the word MONOS IS permuted in all posiblities ways and arranged in alphabetical order.

a) OSMON b) OSNOM c) OSMNO d) ONMSO

Solution-

Arranging in alphabetical order MNOOS

M _____ CAN BE ARRANGED IN 4!/2!=12 WAYS

N _____CAN BE ARRANGED IN 4!/2!=12 WAYS (12+12=24)

O _____CAN BE ARRANGED IN 4!/2!=12 WAYS (12+12+12=36) OUT OF BOUND

OM ___ CAN BE ARRANGED IN 3!=6 WAYS(12+12+6=30)

30TH WORD IS OMSON

31ST WORD IS ONMOS

32ND WORD IS ONMSO

Q 61. One card is lost out of 52 cards. two cards are drawn randomly. They are spade. What is the probability that the lost card is also spade?

a)1/52 b)1/13 c) 1/4 d) 11/50

Solution- (13 - 2) / (52 - 2) = 11 / 50

62. What is the value of

44444445*88888885*44444442*44444438/44444448^2

a)88888883 b)88888884 c)88888888 d)44444443

Solution- take x=44444444

that equ.can be written as- (x+1) (2x-3) (x-2)+(x-6) / x^2

solving equation we get 2x-5

substitute x=44444444 in above equation we get- 88888883

Q 63.Complete the series 4,20,35,49,62,74,?

a)76 b)79 c) 78 d) 85

Solution- 4+16=20

20+15=35

35+14=49

49+13=62

62+12=74

74+11=85

Q 64. Walking at 3/4 of his normal speed, Mike is 16 minutes late in reaching his office. The usual time taken by him to cover the distance between his home and his office is

a). 42 minutes

b). 48 minutes

c). 60 minutes

d). 2 minutes

Solution- Let s = his normal speed

t = his normal time

Then

D = (3/4)s * (t+16)

Since the distance is the same we can equate this to his regular day which is D = s*t

s*t = (3/4)s * (t+16)

t=48

Q 65. The sum of 5 numbers in AP is 30 and the sum of their squares is 190. Which of the following is the third term?

a)5 b) 6 c) 8 d) 9

Solution- consider the 5 numbers in AP as a-2d,a-d,a,a+d,a+2d;

given,a-2d+a-d+a+a+d+a+2d=30;

5a=30==>a=6

here a is the 3 rd term so..third term is 6.

Q 66. A, B and C can do a piece of work in 20, 30 and 60 days respectively. In how many days can A do the work if he is assisted by B and C on every third day?

a). 12 days

b). 15 days

c). 16 days

d). 18 days

solution- A's 2 day's work =2 * (1/20)= 1/10

(A + B + C)'s 1 day's work =((1/20)+(1/30)+(1/60))= 1/10

Work done in 3 days =(1/10)+(1/10)= 1/5

Now, (1/5) work is done in 3 days.

Whole work will be done in (3 x 5) = 15 days.

Q 67. The least number which when divided by 48, 60, 72, 108 and 140 leaves 38, 50, 62, 98 and 130 as remainders respectively, is:

a) 4562

b) 15110

c) 2135

d) 7589

Solution- Here (48 - 38) = (60 - 50) = (72 - 62) = (108 - 98) = (140 - 130) = 10 in every case

Leat number will be LCM of (48, 60, 72, 108, 140) - 10

LCM = 15120

So, required number = 15120 - 10 = 15110

Q 68. A sum of money is borrowed and paid back in two annual instalments of Rs.882 each allowing 5% C.I.The sum borrowed was:

a) Rs.1680 b) Rs.1142 c) Rs.640 d) Rs.1640

Solution-

amount=p*(1+r/100)^n

borrowed amount= 882 / (1+5/100) + 882 / (1+5/100)^2

= (882*20)/21 + (882*20*20)/(21*21)

= 1640

Q 69. how many parallelograms are formed by a set of 4parallel lines intersecting an other set of 7 parallel lines.

a) 125 b)126 c) 127 d) 128

Solution- Let there be 4 horizontal set of parallel lines and 7 vertical set of parallel line.(U can also consider vice-versa)

Now for a parallelogram, u need 2 horizontally parallel and 2 vertically parallel lines ie. we need to choose 2 lines from each set.

So the solution will be 7C2*4C2=126

Q 70. A completes 80% of a work in 20 days. Then B also joins and A and B together finish the remaining work in 3 days. How long does it need for B if he alone completes the work?

a) 37 ½ b) 32 c)32 ½ d) 37

Solution- let total work is 100 unit .

A can do 80 unit in 20 days .... so he can do 4 unit in 1 days .

now A has finished 80 units so remaining work is 20 unit

so , 20 unit work takes 3 days time to complete in which A will do 12 unit

(since A do 4 unit/days)

thus ,B do 8 unit in 3 days

and total work is 100 unit so B will take (3/8)*100 = 37.5 days to complete total work

Q 71. a person starts writing all 4 digits numbers.how many times had he written the digit 2?

a) 4200 b) 4700 c) 3700 d) 3200

Solution-1)when 2 is at unit place=9*10*10*1

2)when 2 is at tenth place=9*10*1*10

3)when 2 is at hundred place=9*1*10*10

4)when 2 is at thousand place=1*10*10*10

so total no. of 2s=900+900+900+1000=3700

Q 72. There is a tank,and two pipes A and B.A can fill the tank in 25 minutes and B can empty the tank in 20 minutes.If both pipes are opened at same time.how much time required the tank can fill?

a) 15 min b) 18 min c) 13 min d) Never be filled

Solution-We have pipe A filling the tank completely in 25 minutes.

Ie, it can fill 100% tank in 25 minutes.

Hence, in 1 min, the tank is filled = 100/25 = 4%

We have pipe B which can empty the tank in 20 minutes.

Hence, in 1 min, tank is empties = 100/20 = 5%.

Since we have the rate of emptying the tank more than the rate of filling the tank,

We can say that the tank will never get filled.

Q 73. An Old man and a Young man are working together in an office and staying together in a near by apartment. The Old Man takes 30 minutes and the Young 20 minutes to walk from apartment to office. If one day the old man started at 10:00AM and the young man at 10:05AM from the apartment to office, when will they meet?

a) 10:15 b) 10:30 c) 10:45 d) 10:00

Solution-Ratio of old man speed to young man speed = 2:3

The distance covered by old man in 5 min = 10

The 10 unit is covered with relative speed=10/(3-2)=10 min

so, they will meet at 10:15 am.

Q 74. the shopkeeper charged 12 rupees for a bunch of chocolate. but i bargained to shopkeeper and got two extra ones, and that made them cost one rupee for dozen less then first asking price .

how many chocolates i recieved in 12 rupees ??

a)10

b)16

c)14

d)18

Solution-let no. of chocolates==x/12dozen

price of x/12 dozen is rs 12

so price of 1 dozen is 144/x

2nd condition

144/(x+2) +1=144/x

go through option x=16

Q 75. there are 16 teams divided in 4 groups.Every team from each group will play with each other once. The top 2 teams will go to the next round and so on the top two teams will play the final match. Minimum hw many matches will be played in that tournament?

a) 43 b) 40 c) 14 d) 50

Solution-for first round 4*4C2=24

second round 2*4C2=12

1*4C2=6

FINAL=1

TOTAL=43

Q 76. A sealed envelope contains a card with a single digit written on it. Three of the following statements are true and one is false.

I. The digit is 1.

II. The digit is not 2.

III. The digit is not 9.

IV. The digit is 8.

Which one of the following must necessarily be correct?

a) II is false

b) III is true

c) IV is false

d) The digit is even.

e) I is true

Solution-III is true

since when 3 is wrong then there are 2 possibilities that 1 or 8

which is not possible

Q 77. Tickets are numbered from 1,2....1100 and one card is drawn randomly what is the probability of having 2 as a digit?

a) 29/110 b) 32/110 c) 30/110 d) 22/110

Solution-for every 100 probability of having 2 as digit is 19 and 2 is used 20 times for every 100.according to qstn, there r 10 100's having 19 digits as 2 n 200 to 299 has 100 digits having 2. hence,

((10*19)+100)/1100=29/110

Q 78. How many 2's are there between the terms 112 to 375?

a) 313 b) 159 c) 156 d) 315

Solution-156

from 112 to 199. .. (19)

from 200 to 299. .. (100+20)

from 300 to 375. .. (18)

so 100 + 19 + 18 + 20 -1=156

-1 bcz 112 shouldn't be taken

Q 79. Ram and Shakil run a race of 2000 meters. First, Ram gives Shakil a start of 200 meters and beats him by one minute. If , Ram gives Shakil a start of 6 minutes Ram is beaten by 1000 meters. Find the time in minutes in which Ram and Shakil can run the races separately.

a) 12,18 b) 10,12 c) 11,18 d) 8,10

Solution-Let x and y are the speeds of Ram and Shakil.. Then by problem we got following equation

2000/x=(1800/y)-1

1000/X=(2000/y)-6

solve equation 1 and 2, we get x=250 and y=200

Therefore, Time taken by Ram and Shakilk to complete a race of 2000m is 8 min and 10 Min

d) 8,10

Q80. The average temperature of June, July and August was 31 degrees. The average temperature of July, August and September was 30 degrees. If the temperature of June was 30 degrees, find the temperature of September (in degrees).

a) 25

b) 26

c) 27

d) 28

Solution-c.27

June+July+August=3*31=93. ..... **(1)**

July+august+Sept=3*30=90. ..... **(2)June=30**

Hence, putting june =30 in equ 1

July+August=63

putting in 2nd equ.

63+Sept=90

hence, Sept=27

Q 81.Three generous friends, each with some money, redistribute the money as follows: Sandra gives enough money to David and Mary to double the amount of money each has. David then gives enough to Sandra and Mary to double their amounts. Finally, Mary gives enough to Sandra and David to double their amounts. If Mary had 11 rupees at the beginning and 17 rupees at the end, what is the total amount that all three friends have?

a) 105 b) 60 c) 88 d) 71

Solution-let sandra, david and mary each has s, d and 11(given) respectively.

After first distribution

David has d+d=2d, marry has 11+11=22 and sandra has s-d-11.

After second distribution,

sandra has 2*(s-d-11) , mary has 2*22=44 and david has 2d-(s-d-11)-22=3d-s-11.

After third distribution,

sandra has 2*2(s-d-11), david has 2*(3d-s-11) and mary has 44-2(s-d-11)-(3d-s-11)=77-s-d

It is given that finally Mary has 17 rs. So,

77-s-d=17

=>s+d=60

=> s+d+11(Mary's money)=60+11=71.

Q 82. George walks 36 kms partly at a speed of 4 kms per hour and partly at 3 km per hour If he had walked at a speed of 3km per hour when he had walked at 4 and 4 km per when he had walked at 3 he would have walked only 34 kms. The time (in hours) spent by George in walking was).

a) 8 b) 12 c) 5 d) 10

Solution-the question can be solved with the help of two equations...

4x+3y=36--- equ(1)

3x+4y=34--- equ(2)

by solving these two equations

x=6 and y=4

so total time is 10hours.

Q 83. The sum of the four consecutive two digit odd number when divided by 10 becomes a perfect square, which of the following can be one of these four numbers?

a) 21 b) 25 c) 41 d) 67

Solution-Let the four 2-digit odd numbers be

n-3 n-1 n+1 n+3

Sum of the 4 numbers ==> 4n

acc to qn, when the sum is divided by 10

we get a perfect square...

perfect squares include==>1,4,9,16,25,36,49,.....

Possible values of 4n/10 ==> 4, 16,36...

If 4n/10=4

n=10

Hence, the corresponding nos are 7,9,11,13(all of which are NOT 2-digit nos)

f 4n/10=16

n=40

Hence the corresponding nos are 37, 39, 41, 43

If 4n/10=36

n=90

Hence the corresponding nos are 87, 89, 91, 93

The answer to the ques therefore is Option C

Q 84. Consider the sequence of numbers 0, 2, 2, 4,... Where for n > 2 the nth term of the sequence is the unit digit of the sum of the previous two terms.

Let sn denote the sum of the first n terms of this sequence. what is the smallest value of n for which sn>2771?

a) 692 b) 693 c) 694 d) 700

Solution-0 2 2 4 6 0 6 6 2 8 0 8 8 6 4 0 4 4 8 2 .0 2 2...this series will be repeated after every 20 terms

sum of 20 terms of series=80

2771/80 quotient=34 remainder=51 sum of 13 terms =52

number of terms is 693 since 34*20=680

680+13 =693

sum of 693 terms will be > 2771

Q 85. a number plate can be formed with two alphabets followed by two digits with no repetition. then how many possible combinations can we get?

a) 52500 b) 58500 c) 56500 d) 56800

Solution-no.of alphabets=26 (a-z), no.of digits=10(0-9).

ways of arranging two alphabets with out repetition=26*25;

ways of forming two digits without repetition=10*9

no.of combinations of forming the number on number plate= 26*25*10*9=58500

Q 86. I bought a certain number of marbles at rate of 59 marbles for rupees 2 times M, where M is an integer. I divided these marbles into two parts of equal numbers, one part of which I sold at the rate of 29 marbles for Rs. M, and the other at a rate 30 marbles for Rs. M. I spent and received an integral number of rupees but bought the least possible number of marbles. How many did I buy?

a) 870 b) 102660 c) 1770 d) 1740

Solution-Let he bought x marbles.

59 marbles costs = Rs. 2M

Therefore, x marbles costs = Rs. (2M * x) / 59

Since the marble is divided into 2 equal parts so the number x should be an even number.

For first x/2 marbles,

29 marbles s.p. is = Rs. M

Therefore, x/2 marbles s.p. = Rs. (M * x) / 58

For other x/2 marbles,

30 marbles s.p. is = Rs. M

Therefore, x/2 marbles s.p. = Rs. (M * x) / 60

Now we can't equate like [(M * x) / 58] + [(M * x) / 60] = (2M * x) / 59

because (M *x) will get cancel each side and of course 1/58 + 1/60 is not equal to 2/59

So here we don't need M and we can cancel it. After that we have,

CP = 2x/59

For first x/2 marbles,

SP = x/58

And for other x/2 marbles,

SP = x/60

Now this CP and SP must be an integer (as per question). So we have to find a number x which will be divisible simultaneously by 59, 58 and 60. So we have to find the LCM of 59, 58, 60 which will turn out minimum value as 102660 and it is even as well. So the value of x will be 102660 minimum

Q 87. Cara, a blue whale participated in a weight loss program at the biggest office. At the end of every month, the decrease in weight from original weight was measured and noted as 1, 2, 6, 21, 86, 445, 2676. While Cara made a steadfast effort, the weighing machine showed an erroneous weight once. What was that.

a) 2676

b) 2

c) 445

d) 86

Solution-1*1+1=2

2*2+2=6

6*3+3=21

21*4+4=88 and not 86

Q 88. How many different integers can be expressed as the sum of three distinct numbers from the set {3,8,13,18,23,28,33,38,43,48}

a) 421 b) 20 c) 10 d) 22

Solution-minimum sum of 3 no.=24

maximum sum =129

now the series 24,29,34,. ..... ,129

no of terms ={(129-24)/5}+1=22

Q 89. Aman walking at the speed of 4 km/h crosses a square field diagonally in 3 minutes. The area of the field (in m2) is:

a) 20000 b) 21000 c) 25000 d) 26000

Solution-convert speed into m/s i.e

4*5/18 = 2/9

this is distance covered in 1 sec.

so distance covered in 3 min i.e 180 sec = 2/9*180

= 200m

this is the lenth of diagonal

and are of square = 1/2 (daigonal)*(diagonal)

1/2*200*200

20,000 m2

Q 90. A owes B Rs 50. He agrees to pay B over a number of consecutive day starting on a Monday, paying single note of Rs 10 or Rs 20 on each day. In how many different ways can A repay B. (Two ways are said to be different if at least one day, a note of a different denomination is given)

a) 8

b) 7

c) 6

d) 5

Solution- 10,20,20=3!/2!==3 ways

10,10,10,20=4!/3!==4 ways

10,10,10,10,10=5!/5!==1 way

so.total=3+4+1==8 ways.

Q 91. The HCF of 2472, 1284 and a third number ‘N’ is 12. If their LCM is 2^3 × 3^2 × 5 × 103 × 107,

then the number ‘N’ is:

a) 2^2 × 3^2 × 7

b) 2^2 × 3^3 × 103

c) 2^2 × 3 × 5

d) None of these

Solution- 2472=8*3*103

1284=4*3*107

so n = 4*3*x

dividing by hcf

2472/12=2*103

1284/12=107

lcm/12=2*3*5*103*107

x=lcm/first*second

x=2*3*5*103*107/2*103*107=3*5=15

number is = 15*12=180

ans. d)

Q 92. A certain function f satisfies the equation f(x)+2*f(6-x) = x for all real numbers x. The value of f(1) is

a) 2

b) can't determine

c) 1

d) 3

Solution- f(1)+2*f(6-1)=1. .....**(1)**

f(5)+2*f(6-5)=5...... **(2)**

substituting we have (2) in (1) we have :-

-3f(1)=-9,

hence answer f(1)=3

Q 93. what is the value of
77!*(77!-2*54!)^3/(77!+54!)^3+54!*(2*77!-54!)^3/(77!+54!)^3

a)2*77!+2*54! b) 77!-54! c) 77!+54! d) 2*77!-2*54!

Solution-77!-54!

take 77!=a, 54!=b then we get in simple (a-2b)^3/a+b solve it u get a-b as answer

Q 94.Find sum of the series 1-2+3-4+....-98+99

a)-49 b) 0 c) 50 d)-50

Solution-(1+3+.....+99)-(2+4+. .. **+98)ie 50 terms+49 terms resp.**

using formla s=n/2[a+l]

we get, 50/2[1+99]-49/2[2+98]

den 2500-2450=50.

Q 95. In a city there are few engineering, MBA and CA candidates. Sum of four times the engineering, three times the MBA and 5 times CA candidates is 3650. Also three times CA is equal to two times MBA and three times engineering is equal to two times CA. In total how many MBA candidates are there in the city?

a) 200 b) 300 c) 450 d) 400

Solution-e = no. of engineering students, m = no. of MBA students and c = no. of CA students

4e + 3m + 5c = 3650, therefore e = (3650-3m-5c)/4 --------(1)

3c = 2m , therefore c = 2m/3

3e = 2c replacing e with (1) we get,

3(3650- 3m - 5c)/4 = 2c. ... replacing c with 2m/3 and solving this equation we get m = 450

So the number of MBA students is 450

Q 96. Find the sum of angles 1,2,3,4,5 in a star.

a) 180 b) 300 c) 360 d) 400

Solution-et an angle of 5 star is a,b,c,d,e

then (180-a)+(180-b)+(180-c)+(180-d)+(180-e)=720

now a+b+c+d+e=900-720=>180

Q 97. Consider a triangle drawn on the X-Y plane with its three vertices at (41, 0), (0, 41) and (0, 0), each vertex being represented by its (X, Y) coordinates. The number of points with integer coordinates inside the triangle (excluding all the points on the boundary) is

(1) 780

(2) 800

(3) 820

(4) 741

Solution-draw the triangle and view it carefully

The number of points with integer coordinates inside the triangle are

1,39)

(1,38),(2,38)

(1,37),(2,37),(3,37)

.....

.....

(1,2),(2,2),(3,2),. ..... **(38,2)**

(1,1),(2,1),(3,1),(4,1). (38,1),(39,1)

so total no of points are=1+2+3+. +38+39

=39*(39+1)/2=39*20=780

ANS(1)

Q 98. the marked price of coat was 40%less than the suggested retail price. Eesha purchased the coat for half of the marked price at the 15th anniversary sale. What percent less than the suggested retail price did eesha pay?

a)60% b)20% c)70% d)30%

Solution-suppose retail price = 100

so the market price will be = 60

as given coat purchased = half f d market price = 30

so its clear isha paid 70% less than retail price.

Q 99. There is a school were 60% are girls and 35% of the girls are poor.

Students are selected at random, what is the probability of selecting a poor girl out of total strength.

a) 21 b) 27 c) 28 d) 29

Solution- '21 will be the probability of poor girl out of total strength....

let 100 be the total strength

60% out of 100= 60 number of girls

now 35% of gals are poor ..i.e. (35/60)*100 are poor girl=21

so 21% of total strength are poo... i.e. the probability is .21

Q 100. If m+n is divided by 12 leaves a reminder 8,If m-n is divided by 12 leaves a reminder 6,then If mn is divided by 6 what is the reminder?

a) 4 b) 3 c) 2 d) 1

Solution- let m=19 and n=1;

m+n=20, gives the remainder=8

m-n=18, remainder=6

so mn=19*1=19 after division by 6 it wud giv the remainder 1...:)

Q 101. there is conical tent in which 10 persons can stand. Each person need 6m square to stand and 60m cube air to breath. what is the height of tent?

a) 60 b) 30 c) 20 d) 45

Solution-Amount of area reqd. by 1 person to stand = 6 sq.metre

Amount of area reqd. by 10 person to stand = 6x10 sq.metre i.e 60 sq.metre

So,

Base Area, pi*r*r=60

=>r*r = 60/pi ---------- (1)

Now,

Volume of air reqd. to breathe 1 person= 60 cu.metre

Volume of air reqd. by 10 person to breathe = 60x10=600 cu.metre

So,

Volume, 1/3*pi*r*r*h = 600----------(2)

Putting the value of eqn 1 in eqn 2, we get

1/3*pi*60/pi*h = 600

==> 1/3*60*h = 600

==> 20*h = 600

==> h = 600/20 = 30

Therefore height = 30 metres

Q 102. In a triangle ABC, the lengths of the sides AB and AC equal 17.5 cm and 9 cm respectively. Let D be a point on the line segment BC such that AD is perpendicular to BC. If AD = 3 cm, then what is the

radius (in cm) of the circle circumscribing the triangle ABC?

a) 17.05 b) 27.85 c) 22.45 d) 26.25

Solution-circum radius = R

Area of the triangle ABC = Δ = ½ (BC)(AD) = ½ (a)(3) = 3a/2

The formula here is : R = abc/4Δ = abc / 4(3a/2) = bc/6 = (17.5)(9)/6 = 26.25

Q 103. A rectangle is divided into four rectangles with area 70,36,20 and X. The value of X is:

a) 350/90 b) 350/7 c) 350/11 d) 350/13

Solution-70/x = 36/20

=> x*36= 20*70

=> x= 20*70/36 = 350/9

Q 104. the ratio of radii of cylinder to that of cone is 1:2.heights are qual.find ratio between volume.

a) 3:4 b) 1:2 c) 1:4 d) 4:1

Solution-Let ratio of radius of cylinder to cone=...r1/r2=1/2

Volume of cylinder=π*r1^2*h

Volume of cone=π*r2^2*h/3

Ratio of volumes=3*(r1/r2)^2= 3:4

Q 105. A hollow pipe has circumference 14 cm. A bug is on its wall at a distance of 48 cm from top. A drop of honey is on the wall at 24 cm from top but diametrically opposite to bug. Find the shortest distance bug has to travel to reach honey.

a) 25 cm b) 39 cm c) 21 cm d) 24 cm

Solution-Total Vertical Distance bug has to through = 24 cm Circumference of pipe= 14 cmSince bug has to move to diametrically opposite side, it has to cover half of the circumference = 14/2 = 7 cmSO by Pythagoras, diagonal distance would be = sqrt(7^2+24^2) = sqrt(625) = 25 cm

Q 106. if a ladder is 100m long, and distance b/w bottom of ladder and wall is 60. top side of bottom and wall is joint.

what is the maximum size of cube that place b/t them.

a) 34.28 b) 24.28 c) 21.42 d) 28.56

Solution- using trigonometry we have

(80/60)=tan(y).....1

(x/60-x)=tan(y).... 2

solve it.....u will have 35 so a) 34.28

Q 107. what is the next three numbers for the given series? 11 23 47 83 131

a) 145 b) 178 c) 176 d) 191

Solution- there are multiples of 12:

11+12=23

23+(12*2)=47

and so on..... answer is 191 as 131+(12*5)=191

Q 108. A series of book was published at 7 year intervals.When the 7th book was issued the sum of publication year is 13524.When was the 1st book published ?

a) 1911 b) 1910 c) 2002 d) 1932

Solution- sum of A.P. series=(n/2)(2*a+(n-1)*d)

here sum=13524,n=7,d=7 substituting these values in above eq. n solving we get

a=1911

which was the 1st book published year.

Q 109. There are 14 digits of credit card number to be filled. Each of the below three boxes contains continuous digits of 18 as sum. Given: 4th digit is 7 and 7th digit is x. Then what is the value of x?

a) 1 b) 7 c) 4 d) 2

Solution- a+b+c=18 and b+c+d=18

given d=7 ,so b+c=11

then we get b=5 and c=6

c+d+e=18, so e=18-(c+d)=5

since d+e+f=18

we get f=7 ie x=7

Q 110. Crusoe hatched from a mysterious egg discovered by Angus, was growing at a fast pace that Angus had to move it from home to the lake. Given the weights of Crusoe in its first weeks of birth as 5, 15, 30,135, 405, 1215, 3645. Find the odd weight out.

a) 3645

b) 135

c) 30

d) 15

Solution- 30.

5

5*3=15

15*3=45(here it is given as 30)

45*3=135

135*3=405

405*3=1215

1215*3=3645

Q 111. Arun makes a popular brand of ice-cream in a rectangular shaped bar 6 cm long, 5 cm wide and 2 cm thick. To cut costs, the company had decided to reduce the volume of the bar by 19%. The thickness will remain the same, but the length and width will be decreased by the same percentage. The new width will be.

a) 4.5cm b) 5.5cm c) 6.5 cm d) 7.5cm

Solution- volume v=l*b*t

new volume is 81% of old v.

therefore new vol = 81%(v)=l'*b'*t

=> 81% (l*b*t) = l'*b'*t

=> 81% (l*b) = l' * b'

since the change in new l and b is the same. hence consider change to be x.

=> 81% (l*b) = xl*xb

81%=x^2

x=9%.

9% change in 5cm width is 0.45

therefore new width is 5-0.45=4.55cm

Q 112. A can complete a piece of work in 8 hours, B can complete in 10 hours and C in 12 hours. If A,B, C start the work together but A laves after 2 hours. Find the time taken by B and C to complete the remaining work.

a) 2 (1⁄11) hours

b) 4 (1⁄11) hours

c) 2 (6⁄11) hours

d) 2 hours

Solution- A,B,C 1 hour work is=1/8+1/10+12= 37/120

A,B,C work together for 2 hours, so,A,B,C 2 hours work is: (37/120)*2=37/60

remaining work=1-37/60=23/60 (23/60 work is done by B and C together)

B,C 1 hour work is= 1/10+1/12=11/60

23/60 hours work done by B,C= (11/60)* (60/23)=11/23

so... ans is: 2(1/11)

Q 113. What is the greatest possible positive integer n if 8^n divides (44)^44 without leaving a remainder?

a)14 b)28 c)29 d)15

Solution- 44^44 = 2^88 *11^44

= 8^29 * 2 * 11^44

so ans is 29

Q 114. A tree of height 36m is on one edge of a road of width 12m. It falls such that the top of the tree touches the other edge of the road. Find the height at which the tree breaks.

a) 16 b) 24 c) 12 d) 18

Solution- let the height at which it broke be Xmt

let the length of other piece be Ymt which touches ground

X+Y=36;

Y^2 - X^2 = 144

then on solving x=16mt

Q 115. How many 6 digit even numbers can be formed from digits 1 2 3 4 5 6 7 so that the digit should not repeat and the second last digit is even?

a)6480 b)320 c)2160 d)720

Solution- given 6th digit even number , so last digit 2 or 4 or 6-> 3 ways

5th digit should be even...so there will be 2 ways(rep. not allowed)

so,therefore we get 5*4*3*2*2*3=720 ways

Q 116. At the end of 1994 rohit was half an old as his grand mother.The sum of years in which they were born is 3844. How old rohit was at the end of 1999.

a) 48 b) 55 c) 49 d) 53

Solution- let at the end of 1994 grand mother's age is x

and rohit's age x/2

then we can say.…birth year of GM is =(1994-x)

and rohit is = (1994 - x/2)

sum of years is 3844

i.e (1994 -x) + (1994 - x/2) = 3884

=> x = 96

i.e GM age is 96

so rohit age will be 96/2 = 48 years

in 1994 age is 48

1995 49

1996 50

1997 51

1998 52

1999 53

so ans should be 53 years.....

Q 117. Find the number of divisors of 1728.?

a) 28 b) 21 c) 24 d) 18

Solution- 1728= 2^6 * 3^3

Hence the Number of factors = (6+1) x (3+1) = 7 x 4 = 28.

We know that if a number represented in standard form (a^m *b^n) , then the number of factors Is given by (m+1)(n+1).

Answer is 28

Q 118. 17 x 8 m rectangular ground is surrounded by 1.5 m width path. Depth of the path is 12 cm. Gravel is filled and find the quantity of gravel required.

a) 5.5

b) 7.5

c) 6.05

d) 10.08

Solution- area of the rectangular ground=(17*8)m^2=136 m^2

taking into account the path:

total area=[(17+(2*1.5))*(8+(2*1.5))]=220 m^2

area of the path=220-136=84m^2

gravel required=[84*(12/100)]=10.08m

Q 119. Ashok, Eesha, Farookh, and Gowri ran a race. Ashok said, “I did not finish 1st or 4th “.

Eesha said, “I did not finish 4th ”. Farookh said, “I finished 1st ”. Gowri said, “I finished 4th ”. There were no ties in the competition, and exactly three of the children told the truth. Who finished 4th? explain

a) Farookh

b) Eesha

c) Gowri

d) Ashok

Solution- Gowri finished with 4th place:

According to the qsn :

exactly three of the children told the truth.

then 4conditions may arise i.e.

TTTF,TTFT,TFTT,FTTT

For 1st condition :

Let Gowri lies that means she nver be in 4th Place and other 3 tell the truth then nobody is in 4th place

so from the above we conclude that Gowri finished at the 4th

Q 120. A circle has 29 points arranged in a clockwise manner numbered from 0 to 28, as shown in the figure below. A bug moves clockwise around the circle according to the following rule. If it is at a point i on the circle, it moves clockwise in 1 second by (1 + r) places, where r is the reminder (possibly 0) when i is divided by 11. Thus if it is at position 5, it moves clockwise in one second by (1 + 5) places to point 11. Similarly if it is at position 28 it moves (1 + 6) or 7 places to point 6 in one second. If it starts at point 23, at what point will it be after 2012 seconds?

a) 1 b) 7 c) 15 d) 20

Solution- after 1st second : (1+23%11 = 1) = 2 places [25]

after 2nd second : (1+25%11 = 3) = 4 places [0]

after 3rd second : (1+0%11 = 0) = 1 place [1]

after 4th second : (1+1%11 = 1) = 2 places [3]

after 5th second : (1+3%11 = 3) = 4 places [7]

fter 6th second : (1+7%11 = 7) = 8 places [15]

after 7th second : (1+15%11 = 4) = 5 places [20]

after 8th second : (1+20%11 = 9) = 10 places [1]

now,for the same pattern from 4th sec to 8th sec will repeat itself (5 sec intervals)..

total time = 2012 secs

first 3 secs out of pattern...so time left 2012 - 3 =2009 secs

now no. of repetitions in the leftover time = 2009/5 = 401. .. remainder = 4

for the next 4 iterations following the similar pattern

the position will be 20..

Q 121. A team won 80% of the games it played. It played 5 more games of which it won 3 and lost 2. Its loss percentage changed to 25%. How many games did it play overall?

a) 20

b) 14

c) 16

d) 25

Solution- ans= 20

if game played=x

then lost game=x/5

now they played 5 more games in which they lost 2

o, (x+5)*25/100=x/5 + 2

x=15

so total game he played=15+5 = 20

Q 122. find the sum of the series given below 1(1!)+2(2!)+3(3!)+.....+2012(2012!)

a) 2013!+1 b) 2013!-1 c) 2012!+1 d) 2013!-1

Solution- let x = 1(1!)+2(2!)+3(3!)+.....+2012(2012!) & y = 1!+2!+3!+.....+2012!

x+y = 2(1!)+3(2!)+....+2013(2012!) = 2!+3!+....+2012!+2013!

x+y+1 = 1!+2!+3!+.....+2012!+2013! = y+2013!

x = 2013! -1

Q 123. there is a circle which circumscribes three unit circle which are tangential to each other.what is the circumference of bigger circle.?

a) pi(4+2sqrt3)/sqrt3 b) pi(6+2sqrt3)/sqrt3 c) pi(3+2sqrt3)/sqrt3 d) pi(6+2sqrt3)/sqrt3

Solution- Just draw according to question,

now, join the center of the smaller circles.

radius of bigger circle= 1+ radius of circle circumscribing the equilateral triangle.

Hence radius of bigger circle = 2/root 3 + 1

Hence circumferance = 2* pie * R

ie. pi(4+2sqrt3)/sqrt3

Q 124. a man starts work on monday and works for 8 days and works for every ninth day as his holiday. His 12th day will be on which day

a) Monday b) Wednesday c) Thursday d) Tuesday

Solution- his work cycle includes 9 days(Mon-Mon=8days and 1 day Tue is holiday)

so to get 12 holiday there should be 12 cycle=12*9=108 days

get remainder by dividing it by 7(108/7 i.e 3)

hence from monday 3rd day is wensday

Q 125. the value of a scooter depreciates in such a way that at the end of each year, is ¾ of its value at the beginning of same year. If the initial value of the scooter is rs40,000. What is the value at the end of 3yrs?

a)23125 b)19000 c)13435 d)16875

Solution- As, it is given that , the cost becomes 3/4 at the end of year.

so, after 3 years ,the price of Scooter=40000*3/4*3/4*3/4= Rs.16875

Q 126. At 12:00 hours jake starts to walk from his house at 6kms an hour. At 13:30 hours, paul follows him from jake's house on his bicycle at 8 kmph. When will jake be 3 kms behind paul?

a) 19:00 b) 18:30 c) 20:00 d) 19:30

Solution- upto 13:30 jake covered 6*1.5=9km

then paul need to be ahead of 3 km.so he go 12km.

their relative speed is 2km per hour.(same direction).

so 12km divided by 2 is 6hrs.

hence from 13:30hrs ,6hrs is added.so answer is 19:30 hours.

ANS:19:30 hrs

Q 127. There are five boxes in a cargo hold. The weight of the first box is 200 kg and theweight of the second box is 20% higher than the weight of the third box, whose weight is 25% higher than the first box's weight. The fourth box at 350 kg

is 30% lighter than the fifth box. Find the difference in the average weight of the four heaviest boxes and the four lightest boxes.

a) 80 kg

b) 75 kg

c) 37.5 kg

d) 116.8 kg

Solution- weight of 1st box=200 kg

so weight of 3rd box=250 kg

weight of 2nd box = 300 kg nd 4th box= 350

thus 5th box=500 kg

avg weight of four heaviest boxes= 1400/4=350

nd lightest box=1100/4=275

diff=75 kg

Q 128. A rectangle of height 100 squares and width 200 squares. Squares is drawn on a graph paper. It is colored square by square from top left corner and moving across in a spiral turning right. Whenever a side of this rectangle or a colored square is reached. Which square is colored last (give its row and column numbers). The bottom right square is on row 100, column 200?

a) 51,150

b) 51, 50

c) 50, 150

d) 50, 50

Solution- For (4, 8) rectangle, ends at (3, 2)

Likewise,

For (2, 4) rectangle, ends at (2, 1)

For (3, 6) rectangle, ends at (2, 5)

For (4, 8) rectangle, ends at (3, 2)

For (5, 10) rectangle, ends at (3, 8)

For (6, 12) rectangle, ends at (4, 3)

For (7, 14) rectangle, ends at (4, 11)

For (8, 16) rectangle, ends at (5, 4)

For (9, 18) rectangle, ends at (5, 14)

For (10, 20) rectangle, ends at (6, 5)

Analyzing the above 10 points:

We can get some idea that,

For Even number of rows, End points column increases 1 by previous column.

For Odd number of rows, End points column difference of the present column and the previous even number column.

For End point row, the value repeats two times from row 2.

In our problem, (100, 200) i.e.

Even number row,

So End point column value must be 50 for 200 columns and End point row value must be 51 for 100 rows.

For even number of rows and columns,

Condition: Column =2*Row

Endpoint (R, C) = {((Row/2) +1), Column/4}

so last square to be coloured is (51, 50)

Q 129. A bag contains six sticks of the following lengths 1 cm, 3 cm, 7 cm, 11 cm and 13 cm. three sticks are drawn at random from the bag. What is the probability that we can form a triangle with those sticks?

a) 11/20

b) 1

c) 1/4

d) 2/5

Solution- A bag contains six sticks of the following lengths 1 cm, 3 cm, 7 cm, 5cm,11 cm and 13 cm. three sticks are drawn at random from the bag. What is the probability that we can form a triangle with those sticks

total possibilities=6c3=20

now for making a triangle,, the sum of any 2 sides must be greater than 3rd side so

3 5 7

3 11 13

5 7 11

7 11 13

5 11 13

5/20=1/4

Q 130. 3 mangoes and 4 apples costs Rs 85. 5 apples and 6 peaches costs Rs. 122. 6 mangoes and 2 peaches cost Rs.114. what is the combined price of 1 apple, 1 peach and 1 mango?

a) 37 Rs

b) 39 Rs

c) 35 Rs

d) 36 Rs

Solution- 3m+4a=85. (i)

5a+6p=122. .. **(ii)**

6m+2p=114.…(iii)

Adding (i) (ii) (iii) we get

9a+9m+8p= 321

9a+9m+9p= 321+p

a+m+p=(321+p)/9 ------(iv) It must have to be a integer number.

To make (iv) as an integer p must have to be either 3 or (3+9) or (3+18)or.....

Let check it out..

if p=3 then

5a+18=122

5a=104

a!= an integer so p=3 is false

If a=12, then

by(ii)

5a=122-72

a=10

by(i)

3m=85=40

a=15

a+m+p= 15+10+12=37.

Q 131. how many number x (x being an integer) with 10<=x<=99 are 18 more than sum of their digits?

a) 12

b) 9

c)18

)10

Solution- Let the no. be 10y+z.

then,

10y+z= y+z+18

=> Y=2

So the no. are 20, 21, 22, 23, 24, 25, 26, 27, 28, 29. Total nos. 10

Q 132. F, G, H, K, L, M and N are 8 people. They need to group into 2 with the condition.

*** F and J must be same group**

*** G and N must be in different groups**

*** H and L must be in same group**

*** M and G are not in the same group**

Correct ordering of groups

a) FJ,KL,MN,GH

b) FH,JL,MN,GK

c) FJ,HL,MN,GK

d) FJ,Hl,MN,GH

Solution- ans is c because c satisfies all the conditions specified

Q 133. In how many ways can the digit of the number 2233558888 be arranged so that the odd digits are placed in the even positions?

a) 900 b) 450 c) 225 d) 360

Solution- odd place combination = 5!/2!*2!=30

even place combination= 6!/2!*4!=15

to total=30*15=450

Q 134. Find the probability that a leap year chosen at random will contain 53 sundays.

a) 2/7

b) 3/7

c) 1/49

d) 1/7

Solution- In leap year,we have 366 days.

366/7=52Weeks+2days

we have already 52 sundays,mondays,etc.

then we have 2 days may be

{sunday,monday} , {monday,tuesday} , {tuesday,wednesday} ,.... {saturday,sunday}

we have 2 possible chances out of 7.

hence ans:2/7.

Q 135. A shop sells chocolates It is used to sell chocolates for Rs.2 each but there were no sales at that price.When it reduced the price all the chocolates sold out enabling the shopkeeper to realize Rs 164.90 from the chocolates alone If the new price was not less than half the original price quoted How many chocolates were sold?

a) 39 b) 97 c) 37 d) 71

Solution- just go on trial and error method until u get u a integer. 164.90/1.7=97

Q 136. The Mean of three numbers is 10 more than than the least number and 15 less than the highest. Their median is 5. Find the sum of the three numbers.

a) 5 b) 20 c) 30 d) 25

Solution- given: x+y+z/3=m

y=5;

m=x+10;x=m-10;

z=m+15

sub in above give eq u will get m= 10;

sub in x, y, z =0,5,25=total=30

Q 136. what is the remainder of (16937^30)/31

a) 1 b) 2 c) 3 d) 6

Solution- 16937=16926+11,now 16926 is completely divisible.. So Wat remains is (11^30)/31

Which is (11^6)^5./31

11^6 gives 4 as remainder..so 4^5/31 is remaining...which gives 1 as remainder.

Q 137. 8+88+888+.....+8888.....8888. There are 21 “8” digits in the last term of the series.

Find the last three digits of the sum.

a) 458 b) 648 c) 658 d) 568

Solution- 21*8=168 carry 16

20*8=160+16=176 carry 17

19*8=152+17=169

the value is 968

Q 138. If x^y denotes x raised to the power y, find last two digits of (1941 ^ 3843)+ (1961 ^4181).

a) 02 b) 82 c) 42 d) 22

Solution- we should use power cycle method

take last 2 digit of 1941 take last 2 digits of 1961

41^1=41 r=1 61^1=61 r=1

41^2=81(take last 2 digits of ans)r=2 61^2=21 r=2

41^3=21 r=3 61^3=81 r=3

41^4=61 r=4 61^4=41 r=4

41^5=01 r=0 61^5=01 r=0

41^6=41 61^6=61

hence power cycle is repeating and it is 5 hence for 1961 it is 5

r-remainder for 1961 powercycle-5

for 1941 powercycle-5 4181mod5=rem1=61

3843mod5=rem 3=21

therefore 21+61=82

Q 139. in a g6 summit being held in London a French, a german, an Italian, a british, a Spanish and a polish diplomat represent their respective countries and participate in a round table conference to strengthen the co operation between these countries. There are exactly 6chairs evenly spaced around a circular table. The chairs are numbered 1 through 6, with successively numbered chairs next to each other and chair number 1 next to chair no6. Each chair is occupied by one of the diplomats. The following conditions apply.

- **Polish sits immediately next to british**
- **German sits immediately next to Italian**

- **rench doesn’t sit immediatedly next to Italian**
- **If Spanish sits immediatedly next to polish, Spanish doesn’t sit next to Italian**

Which of the following seating arrangement of the 6 diploamts in chair 1-6 would not violate the given conditions?

A) French, polish, british, Italian, Spanish, german

B) French, german, Italian, polish, british, Spanish

C) French, german, Italian, Spanish, polish, british

D) French, Spanish, polish, british, german, Italian

a) C b) D c) A d) B

Solution- from the above inference its clear that german sits beside italian but spanish must sit beside polish and polish sits nxt to british. So D matches n hence option b)D

Q 140. Figure shows an equilateral triangle of side of length 5 which is divided into several unit triangles. A valid path is a path from the triangle in the top row to the middle triangle in the bottom row such that the adjacent triangles in our path share a common edge and the path never travels up (from a lower row to a higher row) or revisits a triangle. An example is given below. How many such valid paths are there?

a) 120 b) 16 c) 23 d) 24

Solution- given side length=5 ..... so (n-1)!=(5-1)!=4!

ans is 24

Q 141. A sum of rs 3000 is distributed amongst A, B, and C. A gets 2/3 of what B and C got together and c gets 1/3 of what A and B got together, C's share is?

a) 1200 b) 2250 c) 750 d) 1050

Solution- A+B+C=3000 from data it is clear that A=2/3(B+C) and C=1/3(A+B)

==> 3C=A+B so replace A+B can be replaced by 3c.....

===> 4c=3000 and now c=3000/4===>750... So c gets 750

Q 142. Joke is faster than Paul, Joke and Paul each walk 24 KM. The sum of their speed is 7 Km per hour. And the sum of times taken by them is 14 hours. Then, Jake speed is

a) 7 kmph b) 3 kmph c) 5 kmph d) 4 kmph

Solution- peed=TimedistanceSpeed=Timedistance

let the speed of joke x then speed of paul will be 7-x

24x+247−x=1424x+247−x=14

Try to plugin the values from the options. If Jake speed is 4 the paul is 3.

Q 143. A child is looking for his father. He went 90 meters in the east before turning to his right. He went 20 meters before turning to is right again to look for his father at his uncle's place 30 meters from this point. His father was not there. From there, he went 100 meters to his north before meeting his father in a street. How far did the son meet his father from starting point ?

a) 80 metre

b) 90 metre

c) 100 metre

d) 110 metre

Solution- Clearly, the child moves from A to B 90 metres eastwards upto B, then turns right and moves 20 metre upto C, then turns right and moves upto 30 metre upto D. Finally he turns right and moves upto 100 metre upto E.

So AB = 90 metre, BF = CD = 30 metre,

So, AF = AB - BF = 60 metre

Also DE = 100 metre, DF = BC = 20 metre

So, EF = DE - DF = 80 metre

as we can see in image that triangle AFE is a right angled triangle and we are having two sides, need to calculate third one, so we can apply Pythagoras theorem here

A = AE = sqrt{AF^2 + EF^2}

= sqrt{(60)^2+(80)^2}

= sqrt{3600+6400}

= sqrt{10000} = 100

So from starting point his father was 100 metre away.

Q 144. assume that f(1)=0 and f(m+n)=f(m)+f(n)+4(9mn-1).for all natural no(integer>0)m and n.what is the value of f(17)?

a) 5436 b) 4831 c) 5508 d) 4832

Solution- 4832...

calculate upto f(5)by putting values in the given expression i.e. f(1)=0, f(2)=32, f(3)=100, f(4)=204, f(5)=344. then cal f(7)=f(5+2)orf(3+4)=732 and f(10)=f(5+5)=1584, and now f(17)=f(10+7)=f(10)+f(7)+4(9*15*7 -1).....

f(17)=1584+732+2516=4832

Q 145. The numbers 272738 and 232342, when divided by n, a 2 digit number, leave a remainder of 13 and 17 respectively. Find the sum of the digits of n?

a) 7

b) 8

c) 5

d) 4

Solution- as remainder is 13 & 17 so dividend must be greater than 17

272738 = n*a+13 => 272725=n*a ---(1)

232342 = n*b+17 => 232325=n*b --- (2)

last two digit of (1)&(2) is 25 so n must be 25, no other two digit no. greater than 25 satisfies this

272725=25*10909

232325=25*9293

n=25, sum of digits of n = 2+5 = 7

Q 146. 60 48 38 28 24 20 18 choose odd one.

a) 28 b) 38 c) 60 d) 18

Solution- 28 is odd one.because

coming in reverse order 18+2=20

20+4=24

24+6=30

30+8=38

38+10=48

48+12=60

Q 147. The perimeter of a equilateral triangle and regular hexagon are equal.Find out the ratio of their areas?

a) 3:2

b) 2:3

c) 1:6

d) 6:1

Solution- Given that perimeter of equ.triang and hexagon are equal.consider length of triang

as 'x' and length of hex as 'y'.so the relation is x=2y.Hexagon is made of six equ traingles and formula for area of equ triang is sqrt(3)/4*x^2 and using this we get ratio of areas as 2:3

Q 148. what is the remainder of (32^31^301) when it is divided by 9?

a) 3 b) 5 c) 2 d) 1

Solution- 32^31^301

when 31 divided by 9 gives remainder 5

5 5^2 5^3 all gives the same unit digit 5

so 31^32 gives unit digit 5

same rule applicable to 31^301

when 31 divided by 9 gives remainder 4

4 4^2 4^3 4^4 =4 6 4 6 unit place repeats for every 2 times i,e for even power its uint place is 6 and for odd its 4

as 301 is odd its unit place is 4

so 31^32^301=31^4=5^4=5 is the ans

Q 149. A take 12 hrs to make publication B take 10 hrs to make publication. find the time taken by them to make two publications working independently?

a) 12 hours b) 11 hours

b) 22 hours d) 11 hours 40 minutes

Q 150. Which of the following numbers must be added to 5678 to give a reminder 35 when divided by 460?

a) 980 b) 797 c) 955 d) 618

Solution- ans is 797

If you need of remainder 35, while dividing a number by 460. The Unit place of that number must be 5. So 5678+797= 6475

6475/460 gives reminder 35

Q 151. If a number is divided by 357 the remainder is 5, what will be the remainder if the number is divided by 17?

a) 9 b) 3 c) 5 d) 7

solution- ANS:5 ...to get the original number add the 357 with remainder 5, you will get 362.then divide it by 17 you will get 5 as remainder.

Q 152. A girl entered a store and bought x flowers for y dollars (x and y are integers). When she was about to leave, the clerk said, "If you buy 10 more flowers I will give you all for $2, and you will save 80 cents a dozen". The values of x and y are:

a) (15,1)

b) (10,1)

c) (5,1)

d) Cannot be determined from the given information.

Solution- initially price/dozen= 12y/x;aftr purchasing 10 more--new dozen prices= 2*12/x+10...now.. 12y/x- 24/x+10 must be equal to 80/100 dollars...keep substituting the options.. (5,1) fits in

Q 153. In how many possible ways can write 3240 as a product of 3 positive integers a,b and c

a) 450 b) 420 c) 350 d) 320

Solution- 3240=2*2*2*5*3*3*3*3

so,

no. of ways=8!/(3!*4!)=420.....

Q 154. a and b starts from their house at 10am. they travel from their house on MG road at 20kmph and 40kmph. they meet at T junction at 12:00 pm B reaches the T junction earlier and turns right .Both of them continue travelling till 2:00pm what is distance between a& b at 2:00pm?

a) 160km b) 120km c) 140km d) 150km

Solution- a reaches the junction at 12 after travelling 40 km

but b reaches junction at 11 after travelling 40 km.

at 2 am. a will travel 40 km from junction

at 2 am. b will travel for 3 hrs that is 120 km from junction so it travel 10 km.

distance between them is 120 +40=160 km

Q 155. On door A - It leads to freedom

On door B - It leads to Ghost house

On door C - door B leads to Ghost house

The statement written on one of the doors is wrong.

Identify which door leads to freedom.

a) A

b) B

c) C

d) None

Solution- lets consider door A is wrong then A leads to ghost

on door B written is it leads to ghost

and on door c it is written door B leads to ghost house,henc C leads to freedom

in the same way assume B is wrong,means lead to freedm but on door c written door B leads to ghost house,it is contradiction

last case if we assume C is wrong,the written on door B is also becoming wrong

so our ansr is C lead to freedm which comes from1st case where we assume A is wrong

Q 156. The savings of employee equals income minus expenditure.If the income of A,B,C are in the ratio 1:2:3 and their expense ratio 3:2:1 thenwhat is the order of the employees in increasing order of their size of their savings?

a) A>C>B b) B>A>C c) B>C>A d) C>B>A

Solution-C >B>A

salary of C is highest among all n expenses are least

salary of A is lowest among all n expenses are high

Q 157. A manufacturer of chocolates makes 6 different flavors of chocolates. The chocolates are sold in boxes of 10. How many different boxes of chocolates can be made?

(NOTE: A box is considered different from another only if, regardless of the order, the box contains a different number of chocolates of at least one type)

a) 3003

b) 10^6

c) 3000

d) 6^10

Solution- f n similar articles are to be distributed to r persons, x1+x2+x3......**xr=n each person is eligible to take any number of articles then the total ways are n+r−1Cr−1**

In this case x1+x2+x3...... **6=10**

in such a case the formula for non negative integral solutions is n+r−1Cr−1

Here n =10 and r=6. So total ways are 10+6−1C6−1 = 3003

Q 158. 4 men throw a die each simultaneously. Find the probability that at least 2 people get the same number.

a) 5/18 b)13/18 c) 1/36 d) ½

Solution- 13/18 as 1-(probability of not getting same no.)=1-((6*5*4*3)/(6*6*6*6))

Q 159. How many positive integers less than 500 can be formed using the numbers 1,2,3,and 5 for digits, each digit being used only once.

a) 52 b) 68 c) 66 d) 34

Solution- 3 digits no formed= 3*3*2 (5 cant come at place,no should be less than 500)

=18

2 digit no formed=4*3

=12

1 digit no formed=4

total positive integers=18+12+4

=34

Q 160. In a rectangular region of 300X400 foot, there are 3 ants per square inch. how many ants(approximate value) are there in the square region???

[Gave the hint: 1foot=12inches]

a) 500million

b) 50million'

c) 500000

d) 5million

Solution- 00*400 (foot)

300*400*12 (inches)

300*400*12*3

4320000

approximately 50 million ants

Q 161. The letters in the word ROADIE are permuted in all possible ways and arranged in alphabetic order. Find the word in the 44th rank.

a) AERIOD b) AERDOI c) AERODI d) AEODRI

Solution- A -----=> 5!=120

AD ---- => 4!=24

AED --- => 3!=6

AEI ---=> 3!=6

AEO --- => 3!=6

4+6+6+6=42

AERDIO => 43th

AERDOI => 44th

Q 162. There is a pool of radius X and there is a path way around the pool with a width of 4 feet. Find the radius of the pool if the path area / pool area = 11/25.

a) 12

b) 5

c) 25

d) 29

Solution- (PI.(X+4)^2-PI.X^2)/PI.X^2=11/25

(X+4)^2/X^2-1=11/25

(X+4)^2/X^2=36/25

(X+4)/X=6/5

X=20

Q 163. a certain organization has three committees.only two persons are members of all committees,but every pair of committees have three members in common.what is the least possible no of member of members on any one committee?

a) 4 b) 5 c) 6 d) none of these.

Solution- the least possible no of member of members on any one committee = 4 .. option a)

In all 3 committees, say X,Y,Z, 2 persons say A and B are common.

C is common between X and Y.

D is common between Y and Z.

E is common between X and Z.

so X committee has ABCE. Y committee has ABCD. Z committee has ABDE.

Q 164. In how many ways can 2310 be expressed as product of three factors??

1) 41

2) 56

3) 23

4) 46

Solution- (3^(n-1)+1)/2

2310=2*3*5*7*11

n=5

do this

ans===41

Q 165. In an office, at various times during the day, the boss gives secretary a letter to type , each time putting the letter on top of the secretary's inbox. when there is time, the secretary takes the top letter off the pile and type it. if there are 5 letters in all and boss delivers them in the order 12345, which of the following could not be the order in which secretary types them?

a) 24351

b) 45231

c) 32415

d) 12345

Solution- B) 45231 cant be the order of typing.....

Suppose boss put first time 4 letters in order 1,2,3,4

Then definitely she will strt typing the top one tht is 4th..after completing 4th letter...boss again put the 5th letter on the top.…sho she again choose the top most tht is 5th one and type it. after tht queue will have order 1,2,3

So she should shoosed the top most one tht is 3rd card. but in option it is given 2 which is wrong. check like this in all the options.

Q 166. 6 task and 6 persons. P1 and P2 does not do task T1. T2 is assigned to P3 or P4;. Each person should be assigned with at least 1 task. In how many ways the task can be assigned.

a) 192

b) 360

c) 144

d) 180

Solution- 192

As T1 can be done in 4 ways(p3,p4,p5,p6)

T2 in tw0 ways (p3,p4)

and the remaining in 4! ways

4*2*4!=192

Q 167. How many lattice points are there between (2,0) and (16,203)?

a) 8 b) 10 c) 14 d) 15

Solution- draw a line from origin(0,0) to (14,203)

this line is parallel to the line joining(2,0) and (16,203).

A lattice point is defined as the positive values for x and y coordinates along the line in a graph.

after drawing this line you will come across 8 such points.

so the answer is 8

Q 168. It takes 52 days to complete an agreement deal by a certain number of men.After 17 days,300 men are added and 21 days are reduced.how many men were working initially?

a) 250 b) 150 c) 200 d) None of the above

Soultion- Let x no people works initially.

So the left work after 17 days can be done by x people is in 35, but the same work can be done by x+300 is in 35-21=14 days.

so, 35*(x/(x+300))=14

x=200.

initially 200 people worked

Q 169. four parallel lines are drawn parallel to one side of an equilateral triangle such that it cuts the other two sides at equal intervals.the area of the largest segment thus formed is 27msqr.find the area of the triangle?

a) 100 b) 75 c) 81 d) 54

Solution- Area of Trapezoid= ((a+b)*h))/2(where a and b are length of parallel sides)

Now Since side is divided in 5 parts therefore,length of one interval=a/5

Now use Sin60 and to calculate the height of trapezoid(ie. a root 3/10). and cos60 to calculate the base of the triangle whose hypotenuse is a/5..(i.e. will be a/10);

We get b= a-(a/10+a/10)

b= 4a/5

now equate the area of trapezoid with 27..

Hence area of equilateral triangle is 75 sqm

Q 170. In the sample subtraction problem below, single digits are replaced by letters,

Find the values of 3*A + &*B + 4*C *D = ?

A5C1

3U79

397D

a) 80

b) 99

c) 89

d)96

Solution- I think the answer is 96, let me explain...

from the above question it is clear that there is a printing mistake, for now let us assume b=u;

after solving the subtraction we get the value A=7,d=2,u=5,c=5;

now if we suppose &=a then the ans will be 96(Which is option d)

if &=d then the ans will be 71

if &=c then the ans will be 86

so, from all the assumptions only the first one matches with the options given,

so option d will be our answer, 96

Q 171. two dice are thrown. find the probability of getting a multiple of 3 of 4 as the sum.

a) 5/9 b) 4/9 c) 2/9 d) 1/9

Solution- 20/36=5/9

(1,2),(1,3),(1,5),(2,1)(2,2)(2,4)(2,6)(3,1)(3,3)(3,5)(3,6)(4,2)(4,4)(4,5)(5,1)(5,3)(5,4)(6,2)(6,3)(6,6) as its some is multiple of 3 or 4

Q 172. 2 gears. one with 12 teeth and the other one with 14 teeth are engaged with each other. One tooth in smaller gear and one tooth in bigger gear are marked and initially those 2 marked teeth are in contact with each other. After how many rotations of the smaller gear with the marked teeth in the other gear will again come into contact for the first time?

a) 7

b) 12

c) Data Insufficient

d) 84

Solution- Assume the distance between the teeth is 1 cm. Then the circumference of first gear is 12 cm and the second is 14 cm.

Now LCM (12, 14) = 84.

So to cover 84 cm, the first gear has to rotate 84/12 = 7 rounds

(the second gear rotates 84 / 14 = 6 rounds as it is bigger)

Q 173. there are 4 couples who go for honeymoon together.at one of the places they all have to cross the river but only one boat is available.wives are jealous that they dont want their husbands to travel with another women and husbands

are also possessive that they dont want their wives to travel with some other men.the no of minimum possible ways in which they will cross the river are.

a) 16 b) 17 c) 18 d) 19

Solution- we have 4 couple

1) first two husbands goes to another side

2) then one husband comes back

3) then one couple will go

4) then second husband comes back

so in this 4 ways 1 couple reach to another side

so for 4 couples there are 16 ways

a)16 is ans

Q 174. Professor absentminded has a very peculiar problem, in that he cannot remember numbers larger than 15. However, he tells his wife, I can remember any number up to 100 by remembering the three numbers obtained as remainders when the number is divided by 3, 5 and 7 respectively. For example (2,2,3) is 17. Professor remembers that he had (1,1,6) rupees in the purse, and he paid (2,0,6) rupees to the servant. How much money is left in the purse?

option

a) 59

b) 61

c) 49

d) 56

Solution- He had 76 rupees,according to questions, he remember any no. larger than 15 as remainder when divided by 3,5 and 7 respectively.

Divide 76 by 3,5 and 7, we will get (1,1,6)as remainder.

e paid 20 rupees bcoz when 20 divided by 3,5 & 7 , we will get (2,0,6) as remainder.

Now money left=76-20

=56 (Ans)

Q 175. A and B completed a work together in 5 days. Had A worked at twice the speed and B at half the speed, it would have taken them four days to complete the job. How much time would it take for A alone to do the work?

a) 5days b) 20days c) 10days d) 25days

Solution- a nd b's 1 day work, a+b=1/5

with twice d speed of a nd half of b completes work in 4 days,

so, 2a+b/2=1/4.

on solving both d eq. we get a=1/10

so a will complete d whole work in 10 days.

Q 176. 0>a>b>c>d. Which is largest

a) (b+d)/(a+c)

b) (a+d)/(b+c)

c) (b+c)/(a+d)

d) (c+d)/(a+d)

Solution- given statement is 0>a>b>c>d

that means all the values of a,b,c,d are less than ZERO

so lets consider a=-1 b=-2, c=-3 and d=-4 so that 0>a>b>c>d will satisfy

by solving the options we get the values as follows

a.(b+d)/(a+c)= 1.5

b. 1

c.1

d.1.4

among all of these option a is highest so the ans is option a.

Q 177. An engineer undertakes a project to build a road of 15km in 300 days and employs 45 men for the purpose .After 100 days,he finds only 2.5km of the road has been completed.find the (approx)number of extra men must employs to finish the work in time.

option

a) 43

b) 45

c) 55

d) 68

Solution- 45 workers working already

Let x be the total men required to finish the task in next 200 days

2.5 km done hence remaining is 12.5 km

Also, work has to be completed in next 200 days (300 - 100 = 200)

We know that, proportion of men to distance is direct proportion

and, proportion of men to days is inverse proportion

Hence, X = (45 * 12.5 * 100) / (2.5 * 200)

thus, X = 112.5 that is approximately 113

Thus, more men needed to finish the task = 113-45=68

Q 178. there are 5 sweets - jamun, kulfi, peda. laddu and jilebi that i wis t eat on 5 consecutive days, monday throufh friday, one sweet a day, based on following self imposed constraints:

1) laddu is not eaten on monday

2) if jamun is eaten on monady , then laddu must be eaten on friday

3) if jamun is eaten on tuesday, kulfi should be eaten on monday

4) peda is eaten the day following the day eating jelabi

based on the above , peda can be eaten on any day expect??

Solution-monday bcoz the day before jalebi is required to be taken bt its sunday and no sweet is eaten on Sunday

Q 179. A circular swimming pool is surrounded by a concrete wall 4ft wide. If the area of the wall is 11/25 of the area of the pool then radius of the pool in feet is

a) 20

b) 8

c) 16

d) 30

Solution- i thnk ans is 20.Bcoz let r be the radius of pool.

(11/25)(pi*r^2)=(pi(r+4)^2)-(pi*r^2)

By solving this we get ans as 20

Q 180. How many 5's will be there in the number 121122123... till 356?

a) 51 b) 54 c) 50 d) 49

Solution- 121122123... till 356

121,122,123,124... till 356

121 to 200 => 125,135,145, from 150 to 159 => 11 5's , 165,175,185,195

[from 150 to 159, each no. has 1 no. of 5 except 155, 155 has 2 no. of 5]

total=18

201 to 300 => 205,215,225,235,245, 250 to 259 => 11 5's , 265,275,285,295

=> total=20

301 to 356 => 305,315,325,335,345,350,351,352,353,354,355(2 5's),356

=> total=13

total 5's = 18+20+13 = 51

Q 181. Car A leaves city C at 5 pm and drives at a speed of 40 kmph. 2 hours later another car B leaves city C and drives in the same direction as car A. In how much time will car B be 9 km ahead of car A. Speed of car B is 60 kmph.

option

a) 4.25 hrs

b) 4.17 hrs

c) 4.30 hrs

d) 4.45 hrs

Solution-we take the relative speeds of the 2 cars.

relative speed=(60-40)=20km/h

the car a has travelled (40*2)km in 2 hrs.

thence car b has to travel 80km+9km ahead to get ahead by 9 km.

so time= 89/20=4hr 45 min.

Q 182. The rupee/coin changing machine at a bank has a flaw. It gives 10 ten rupee notes if you put a 100 rupee note and 10 one rupee coins if you insert a 10 rupee note but gives 10 hundred rupee notes when you put a one rupee coin!

Sivaji, after being ruined by his rivals in business is left with a one rupee coin and discovers the flaw in the machine by accident. By using the machine repeatedly, which of the following amounts is a valid amount that Sivaji can have when he

gets tired and stops at some stage (assume that the machine has an infinite supply of notes and coins):

a) 26975

b) 53947

c) 18980

d) 33966

Solution-initially sivaji had only one coin so he earns only 1000/-

to earn more he must convert one 100 note to 10 notes and then one ten note to ten coins

now he has Rs.990 and 10 coins

after converting 10 coins he has 10990

after converting another 10 coins he has 20980

after converting another 10 coins he has 30970

after converting another 10 coins he has 40960

after converting another 10 coins he has 50950

after converting another 10 rupee note to coins and using only 3

he has seven coins and 53940

so sum is 53940+7=53947

Q 183. 2/3rd of the balls in a bag are blue, the rest are pink. if 5/9th of the blue balls and 7/8th of the pink balls are defective, find the total number of balls in the bag given that the number of non defective balls is 146.

a) 216 b) 649 c) 432 d) 578

Solution-let total no of balls =x

blue=2x/3

pink=x/3

total no of defective balls = 10x/27 +7x/24

=143x/216

non defective balls=x-143x/216=146 ,x=432

Q 184. One day Eesha started 30 min late from home and reached her office 50 min late while driving 25% slower than her usual speed. How much time in min does eesha usually take to reach her office from home?

a) 80min b) 50min c) 60min d) 70min

Solution-let time taken by eesha daily = x

and usual speed = y

(x+20)*(.75*y) = x*y extra time taken = 50-30 = 20min

.75x+15 = x

.25x = 15

x = 60 min

Q 185. 7,17,19,43,91,131...find the odd term.

a) 17 b) 43 c) 91 d) 131

Solution-except 91 all other are prime no.

so 91 odd one

Q 186. find the no. of zeros in the product of 1^1*2^2*3^3. .. *49^49?

a) 250 b) 225 c) 545 d) 135

Solution-the number of zero's can be found by finding number of 2's and 5's.

since in this number, number of occurrences 2's will always be greater than number of occurrences of 5's.

therefore we need to find number of 5's only.

number of occurrences of 5's

in 5^5 = 5

10^10=10

15^15=15

20^20=20

25^25=25

30^30=30

35^35=35

40^40=40

45^45=45

sum=225.

therefore 225 5's can be paired with 225 2's.

total number of zeros = 225

Q 187. a merchant buys 20kg of wheat at 30rs/kg and 40k at 25rs/kg.he mixes and sells 1/3 rd of mixture at 26rs/kg.price at whch merchant should sell remaining mixture so that a profit of 25% on whole outlay is?

a) Rs30 b) Rs40 c) Rs360 d) Rs 37

Solution-cp of total mixture=(20*30)+(25*40)=1600rs

as he needs 25% profit, so he needs to earn 1600+(1600*(25/100))=2000rs.

he has alredy got rs by selling 1/3 part=1/3 of 60 kg..=20 kg

=20*26=520rs..

now he needs 2000-520=1480rs.

so he must sell remaining 40 kg at=1480/40=37rs/kg

Q 188. the addition of 641+852+973=2456 is incorrect. What is the largest digit that can be changed to make the addition correct?

a) 5 b) 6 c) 4 d) 7

Solution-641

852

973

2466

Given Sum is 2456

So if we change 10's digit 7 to 6 then the sum will be correct .

so 7 is the largest digit .

Q 189. a, b, c are non negative integers such that 28a+30b+31c=365. Then a+b+c is?

a) 13 b)> 13 c) =13 d) 12

Solution-Here, a + b + c = 12

Explanation :

a = 1, because in a month of twelve only Feb has 28 days

b = 4, there are four months namely April,June,Sep and Nov having 30 days

c = 7, there are 7 months namely Jan,Mar,May,Jul,Aug,Oct and Dec having 31 days

So, a + b + c = 1 + 4 + 7 = 12.

Q 190. 26ab5 is a four-digit number divisible by 25. If the number formed from the two digits ab is a multiple of 13, then ab =?

a) 52

b) 65

c) 10

d) 25

Solution-Any number last two digits are 25 then that should be divisible by 25.

When ab=52, the number is 26525 and 52/13=4,

So ab=52

Q 191. A owes B Rs 50. He agrees to pay B over a number of consecutive day starting on a Monday, paying single note of Rs 10 or Rs 20 on each day. In how many different ways can A repay B. (Two ways are said to be different if at least one day, a note of a different denomination is given)

a) 8

b) 7

c) 6

d) 5

Solution-10,20,20=3!/2!==3 ways

10,10,10,20=4!/3!==4 ways

10,10,10,10,10=5!/5!==1 way

so.total=3+4+1==8 ways.

Q 192. After 6 years Raju's fathers age will twice of the Raju's age 2 years ago.His mothers age was twice that of Raju's age. Sum of the age of their parents.

a) 4 less than four times Raju’s age

b) 2 more than four times Raju’s age

c) 4 more than four times Raju's age

d) 2 less than four times Raju's age

Solution-F+6=2(R+6)

F= 2R+6

M-2=2(R-2)

M= 2R-2

Therefore the sum of Raju's Parent's age is

F+M=2R+6+2R-2

F+M=4R+4

4 more than four times Raju's age

Q 193. Length, Breadth and Height of a 3D figure is in the ratio 3:2:1. If the length is doubled and Breadth & Height are halved, then what is the % decrease in the volume of the solid?

a) Decreased by 15% b) Decreased by 18% c) Decreased by 30%

d) Decreased by 50%

Solution-50% decrease

If original length, breadth & height are 3x, 2x and x respectively, then volume=3x*2x*x=6x^3 With length doubled, breadth & height halved, new dimensions are 6x, x and x/2 respectively and volume=6x*x*x/2=3x^3

So % decrease in volume=100*(6x^3-3x^3)/6x^3=50

Q 194. 12 divides, ab313ab (in decimal notation, where a,b are digits>0, the smallest value of a+b is

a) 7

b) 6

c) 2

d) 4

Solution-If a number is divisible by 12 then it should be divisible by 4&3

for divisible by 4 last [2 digit]no's should be divisible by 4

so last no's are 12,15,18,,,,,,and soon..

now least is 12 it,. 3 which is not the option,,

no other least is 16 it's sum 1+6=7..

so 7 is option :)

Q 195. In a telecom assembly factory,there are 250men and 150 women.The average productivity of all works is 12 units per day.The average productivity of men is 15 units per day..what is the average productivity of women per day?

a) 6 b) 9 c) 7 d) 8

Solution-250M + 150W = 400

Thus, 400*12 = 4800

Now, 250*15 = 3750

Now, 4800-3750 = 1050 Thus, for 150W avg prod is : 1050/150 = 7

Q 196. If a lemon and an apple together cost Rs. 12.00, a tomato and a lemon cost Rs. 4.00 and an apple cost Rs.8.00 more than a tomato or a lemon then which of the following can be.

a) Rs 2 b) Rs 4 c) Rs 1 d) Rs 3

Solution-lemon+apple=12rs

tomato+lemon=4

then apple=8+t or 8+l

by solving we get

l+(8+l)=12

2l=4

lemon=2

so a=8+2=10; so a)

Q 197. George, Paul and Hari start a business by contributing Rs. 30000/-, Rs. 40000/- and Rs. 50000/- respectively. After ½ a year George withdraws half his contribution. At the end of the year the business showed a profit of Rs 90000 which was divided amongst the 3 men proportionate the to amount and duration of their investment in the enterprise. Paul got,

a) Rs 25000/-

b) Rs 18000/-

c) Rs 32000/-

d) Rs 24000/

Solution-G,P & h ratio will be

(30,000*6+15000*6):(40000*12):(50000*12)

=27:48:60

hence 135x=90000 x=2000/3

paul got x*48 means 2000/3*48=32000

Q 198. A drinks machine offers three solutions – Tea, Coffee or Random but the machine has been wired up wrongly so that each button does not give what it claims. If each drink costs Rs. 50, what is the mimimum amount of money that must be spent to determine with certainity the correct labeling of the buttons?

a) Rs. 100

b) Can not be determined from the given information

c) Rs 150

d) Rs 50

olution-You have to put just 50rs.

Put 50rs and push the button for Random. There are only 2

possibilities. It will give either Tea or Coffee.

If it gives Tea, then the button named Random is for Tea.

The button named Coffee is for Random selection. And the

button named Tea is for Coffee.

If it gives Coffee, then the button named Random is for

Coffee. The button named Tea is for Random selection. And

the button named Coffee is for Tea.

Thus, you can make out which button is for what by putting

just 50rs and pressing Random selection first.

Q 199. P,Q,R,S are distinct integers numbered from 1 to 12.What is the possible smallest value for (P/Q)+(R/S)=?

a) 1/12+2/11 b) 1/11+9/10 c) 1/11+2/12 d) 1/10+1/11

Solution-he given options for the bove question are, a.1/12+2/11 b.1/11+9/10 c.1/11+1/12 d.1/10+1/11

In mathematics, two things are called distinct if they are not equal.

so p,q,r,s are not equal.

Ans:- 1/12 + 2/11

Q 200. If ab64ab is divisible by 12, then what is the least possible value of a+b ?

a) 4 b) 5 c) 6 d) 7

Solution-1.Condition for divisibility by 3--

The sum of digits of number must be divisible by 3.

i.e a+b+6+4+a+b = 0 mod 3

=> 10 + 2(a+b) = 0 mod 3

=> 1 + 2(a+b) = 0 mod 3 (10 = 1 mod 3)

=> 2(a+b) = -1 mod 3

=> 2(a+b) = 2 mod 3

=> a+b = 1 mod 3

So, a+b must be of the form 3n + 1 for some positive integer n.

Max value of (a+b) is 18 as a and b can take values upto 9 since they are digits.

The possible values are 1,4,7,10,13,16.

2. Condition for divisibility by 4--

For divisibility by 4, Last two digits must be divisible by 4.

i.e 10a+b = 0 mod 4

=> 2a + b = 0 mod 4 (Since, 8a = 0 mod 4)

So, 2a + b can be 4,8,12,16,20,24,28.

Combining these,we get

a + b = 3n +1

2a + b = 4k

The smallest values which satisfies these two equation is a = 4 ,b = 0.

So, the minimum sum of a + b is 4.

Q 201. There are 100 in a class and they attend a test. 20 students are failed in both the subjects. 50 students pass in subject A. 60 students passed in subject B. How many students passed in subject A only.

a) 20 b) 30 c) 15 d) 25

Solution-LET x be the total no. of students who passed in both test.

Then, no. of students who passed in A only= total no. of students who passed in A - total no. of students who passed for both A & B =50 - x

similarly, no. of students who got pass only in B = 60 - x

now, total no. students who got pass = 80

80= (50 - x) +(60-x)+x then we'll get on solving, x=30

students who passed in A only = 50 -x= 50-30=20

Q 202. Aravind can do a job in 24 days. Mani can dig the same well in 36 days. Aravind, Mani, and Hari can do the work together in 8 days. How long does it take Hari to do the work alone?

a) 12 days b) 18 days c) 16 days d) 24 days

Solution-This can be solved with the help of efficiency.Efficiency of (aravind +mani+hari) =total efficiency

Here,

No. Of Days taken by Aravind =24 Efficiency of aravind =1/24

No.of days taken by mani=36 Efficiency of mani=1/36

No. Of days taken when all the three work together =8

Efficiency of all thr three together=1/8

Let the no. Of days taken by hari to complete the job be x.

Then efficiency of hari =1/x Now, from the above formula

1/24 + 1/36 + 1/x = 1/8

x=18

Q 203. Truck A and Truck B move grain into a box at the rate of 20 kilos/min and 13 1/3 kilos a min respectively while truck C removes grain from the box at the

rate of 10 kilos/min. If the capacity of the box is 2.4 tons, and Truck A, Truck B and Truck C are working simultaneously Then the box will be filled in?

a) 1 1/2 hrs

b) 3/5 hrs

c) 1 5/7 hrs

d) 2 1/8 hrs

Solution-otal filled in i min=(20+40/3-10)=70/3

to fill 70/3 time req---- =1min

to fill 2400 time req ---- =(3/70)*(2400/60)hr

=15/7

Q 204. If 5 + 3 + 2 = 151022, 9 + 2 + 4 = 183652, 8 + 6 + 3 = 482466 and 5 + 4 + 5 = 202541, then 7 + 2 + 5 = ?

a) 143547 b) 132234

c) 2577224 d) 112321

Solution-5+3+2=(5*3)(5*2)(5*3+5*2-3(ie second term))=15,10,22

9+2+4=(9*2)(9*4)(9*2+9*4-2(ie second term))=18,36,52

8+6+3=(8*6)(8*3)(8*6+8*3-6(ie second term))=48,24,66

so answer is 143547

Q 205. Two full tanks one shaped like the cylinder and the other like a cone contain liquid fuel the cylindrical tank held 500 lts more then the conolical tank After 200 lts of fuel is pumped out from each tank the cylindrical tank now contains twice the amount of fuel in the canonical tank How many lts of fuel did the cylindrical tank have when it was full?

a) 1200 b) 1000 c) 700 d) 1100

Solution-let area of cylinder=A

& area of cone=B

acc. to ques,

A=B+500..... **(1)**

A-200=2(B-200)..... **(2)**

from (1) &(2)

A=1200

volume of fuel when cylinder is full=1200 lts

Q 206. Asha and Eesha – Eesha lies on Monday, Tuesday and Wednesday. Asha lies on Thursday, Friday and Saturday. Other days they will say the truth. Professor forgot and asked them what day it is. Both of them said yesterday I was lying and then professor got the day. What day it is?

a) Tuesday b) Thursday c) Friday d) Cannot be determined

Solution-Thursday. Asha lies on thursday and today she is lying that she was lying yesterday. Eesha is not lying today but she was lying yesterday.

Q 207. Three sisters are identical triplets. The oldest by is Aasha, and she always tells anyone the truth. The next oldest is Usha, and Usha always will tell anyone a lie. Eesha is the youngest of the three. She sometimes lies and sometimes tells the truth.

Mukund an old friend of the family, visited them one day and was able to recognize who was who, so he asked each of them one question.

Mukund asked the sister that was sitting on the left. “Which sister is in the middle of you three ?” and the answer he received was, “Oh,that’s Aasha.”

Mukund then asked the sister in the middle. “What is your name?” The response given was, “I’m Eesha.”

ukund turned to the sister on the right,then asked. “Who is that in the middle ?” The sister then replied, “She is Usha.”

This confused Mukund.

Who was in the middle ?

a) Aasha b) Eesha c) Usha d) Cannot be determined

Solution-its USHA ! as the left one said the middle one is aasha.. if it was aasha she would have admitted! so the left one and the middle one could be eesha usha both. as eesha sometimes says truth and sometimes lie. and usha always lie. so the right one would be aasha who will say truth and she said the middle one is USHA,so option (c) Usha is a right answer !

Q 208. If 4 examiners can examine a certain number of answer books in 8 days by working 5 hours a day, for how many hours a day would 2 examiners have to work in order to examine twice the number of answer books in 20 days.

a) 6 hours

b) 7 hours

c) 15/2 hours

d) 8 hours

Solution-4 examiner working for 8 days(5 hours a day)

thus,

1 examiner=8*5=40 hrs/person

Now,

2 examiner working for 20 days(x hrs a day) to have twice the work.

thus,2*x hrs spent each day

We need 160 hrs.Working 2 hrs the duration will be:

160/2*x=20

=>x=4 hrs

Twice work is done,thus,time=4*2=8 hrs.

Q 209. A circle has 11 points arranged in a clockwise manner from 0 to 10. A bug moves clockwise on the circle according to following rule. If it is at a point i on the circle, it moves clockwise in 1 sec by (1 + r) places, where r is the remainder (possibly 0) when i is divided by 2. If it starts at 4th position, at what position will it be after 2012 seconds?

a) 7 b) 9 c) 5 d) 1

solution-(D) 1

cycle= 1 +7n after this cycle always is 5th position.

now after 2008 sec point is 5th position.

now next 4 sec 1st position.

or

Trick:- starting position - 3 = present position. maximum cases.

Q 210. in a group of 5, Anooj said"one of us are lying", Pooja said"Exactly two are lying", Bitto said "Exactly three are lying", Billa said"Exactly four are lying", Chitra said,"Exactly five of us are lying". Which one said the Truth?

a) Billa b) Anooj c) Chitra d) Pooja

Solution-Billa because 4 of them are lying and one is saying the truth....

1/5+2/5+3/5+4/5+5/5

15/5=3 So Billa

Q 211. A travels at 40kmph. B travels at 60kmph. They are travelling towards each other. BY the time they meet ,B would have travelled 120 km more than A. Find the total distance.

a)600km b) 720km c) 400km d) 540km

Solution- dist. traveeld by A = 40x (let x be the time taken to meet)

dist travelld by B = 60x

difference in dist = 20x = 120; which gives x=6hours

so relative speed of A and B is 100km/hr.

distance travlld = 6*100 = 600km.

Q 212. If all the numbers between 11 and 100 are written on a piece of paper. How many times will the number 4 be used?

a) 20 b) 19 c) 9 d) none of these

Solution- 19, every word can only be appeared 20 times from 1 to 100. nd it is starting from 11, so 19 times used..

Q 213. 10 people are there, they are shaking hands together, how many hand shakes possible, if they are in no pair of cyclic sequence.

a) 45

b) 9

c) 12

d) 10

Solution- 10c2= n!/(n-r)!.r!

=10!/8!.2!

=10*9/2

=45

Q 214. What should be the value of a,in the polynomials x^2-11x+a and x2-14x+2a so that these two polynomials have common facors.

a) 24 b) 1 c) -1 d) ½

Solution- a=24, then polynomials will become

x^2-11x+24 and x^2-14x+48 which can be written as

(x-8)*(x-3) and (x-8)*(x-6)

(x-8) is common factor.

Q 215. a man cannot remember the number larger than 15. However he remember any number upto 100 by remembering the three numbers obtained as remainders when the numbers is divided by 3,7 and 11 respectively. he remembers that he had (2,4,8) rupees in his purse. and he paid (2,5,4)rupees to the servant. How much money is left in the purse?

a) 40 b) 48 c) 46 d) 52

Solution- the number that satisfies the all remainders of 2,4,8 by numbers 3,7,11 respectively is 74.

and the the number that satisfies the all remainders of 2,5,4 by numbers 3,7,11 is 26.

hence the remaining money is=74-26=48

ans is 48

Q 216. Next number in the given series 1, 7, 8, 49, 50, 56, 57, 343

a) 344 b) 350 c) 2401 d) Cannot be determined

Solution- 344 because

1*7=7

7+1-8

7*7=49

49+1=50

8*7=56

56+1=57

49*7=343

343+1=344

ans= 344

Q 217. A man sold 12 candies in 10$ had loss of b% then again sold 12 candies at 12$ had profit of b% find the value of b.

a) 9 b) 9.09 c) 10 d) 11

Solution- Let the CP be 'x' then:

(x-10)*100/x = b% ----------->1 (12- x) * 100 /x = b% ------->2

(x-10)*100 / x = (12-x)*100 / x

x-10 = 12-x

x = 11

So by Submitting value of x into one of above question (1 or 2) we get b%

which is 9.09 or 9 Approx

Q 218. Two breakers are kept on a table. the capacity of the first breaker is x liters and that of the second breaker is 2x. Two thirds of the 1st breaker and one fourth of the 2nd breaker is filled with wine. The remaining space in both breakers is filled with water. If the content in these breakers are mixed in a larger breaker of volume 3x, What is the proportion of wine in the breaker?

a) 11/12

b) 11/36

c) 7/6

d) 7/18

Solution- 2/3+1/2(1/4th of 2x =1/2 of x)=7/6. when pored in breaker of vol 3x then d proportion will be 7/6 *1/3=7/18

Q 219. When all possible six-letter arrangements of the letters of the word "MASTER" are sorted in alphabetical order, what will be the 49th word?

a) AREMST b) ARMEST c) AMERST d) ARMSET

Solution- ARRANGING THE LETTERS IN MASTER ALPHABETICALLY WE GET.... AEMRST

NOW A E_____can be filled in 4! ways

A M _____ can be filled in 4! ways

so words upto 4!+4! =48 words are of ae series and am series

So 49th word can be A R E M S T

Q 220. find a number such that when it is added to 7249 will be perfectly divisible by 12,14,21,33 and 54

a) 8136 b) 9123 c) 8727 d) 9383

Solution- lcm of 12,14,21,33,54=8316

now checking through options,

on adding 7249+8136/8316=not perfetily divisible

similarly on checking through options,(9383+7249)/8316=2

answer=9383

Q 221.60 men can complete a piece of work in 40 days.60 men start the work but after every 5 days 5 people leave.in how many days will the work be completed?

a) 60 b) 80 c) 120 d) None of these

Solution- (60*40) men complete the work in 1 day

1man's 1 day work is 1/60*40

60 men's 5 day work is 5*(1/40)=1/8

remaining work (1-1/8)=7/8

as 5 men leave

55 men's 1 day work=55*(1/60*40)=11/60*8

now

11/60*8 work done by them 1day

7/8 work done by them [(60*8)/11]*(7/8)=420/11=38.18

so they complete in (38.18+5) days{5 day added bcoz alrdy 5dy work had been done}

43.18 day total work will be completed.

Q 222. A person walks at 4km/hr for a particular duration T1 and 3km/hr for another duration T2 and covers a total distance of 36km. If he walks at 4km/hr for the duration T2 and at 3Km/hr for the duration T1, then he covers only 34km. What willl be the time taken by him to cover the one of the legs?

a) 4hrs

b) 7hrs

c) 10hrs

d) 6hrs

Solution- 4T1 + 3T2 =36...(i)

3T1 + 4T2 =34...(ii)

4*(i)-3*(ii),we get

T1=6hrs and T2=4hrs

Q 223. If ABERSU are in sorted in alphabetical order, if 24 sortings are req for ABUSRE, 25 - AEBRSU, 49- ARBESU, den how many for AEUSRB.

a) 45 b) 48 c) 47 d) 46

Solution- AEUSRB

ALPHA OREDER ABERSU

A B _____ =4! =24

A E B ____ =3!=6

A E R ____ =3!=6

A E S ____ =3! =6

A E U B ___=2! =2

A E U R __ =2!=2

A E U S B _=1

A E U S R B=1

24+6+6+6+2+2+1+1=48

Q 224. a,band c can do some work in 36 days a and b together do twice as much work as c.a and c together do thrice as much work as b.eind time taken by c alone to do complete work

a) 72

c) 120

c) 96

d) 108

Solution- 1 day work for (A+B+C) = 1/36..... (ii)

Now, (A+B) = 2C.....(i)

From (i)and(ii): 3C = 1/36

= 1/108

Thus, C takes 108 days to complete the whole work alone.

(Remaining data are not required in this case)

Q 225. A sum of rs 3000 is distributed amongst A, B, and C. A gets 2/3 of what B and C got together and c gets 1/3 of what A and B got together, C's share is?

a) 1200 b) 2250 c) 750 d) 1050

Solution- A+B+C=3000 from data it is clear that A=2/3(B+C) and C=1/3(A+B)

==> 3C=A+B so replace A+B can be replaced by 3c.....

===> 4c=3000 and now c=3000/4===>750... So c gets 750

Q 226. raj writes a number. He sees that the first no of 2digits exceeds 4times the sum of its digits by 3. If the number is increased by 18, the result is the same as the number formed by reversing its digits. Find the number

a) 35 b) 57 c) 42 d) 49

Solution- 35-4*8=3

35+18=53

answer is 35

Q 227. P (x) = (x2012 + x2011 + x2010 +..... + x + 1)2 – x2012

Q (x) = x2011 + x 2010 + ...+ x + 1

The reminder when P (x) is divided by Q (x) is

a) 1 b) 0 c) x+1 d) x-1

Solution- substitute 1 in x

then it will become

((2013)^2-1)/2012

(2013+1)(2013-1)/2012

2014 is quotient

So, remainder is '0'.

Q 228. If YWUSQ is 25 - 23 - 21 - 19 - 17 then MKIGF is what?

a) 13-11-9-7-6 b) 1-2-3-5-7 c) 9-8-7-6-5 d) 7-8-4-5-3

Solution- YWUSQ - 25-23-19-17

THEN Y=25,W=23,U=21.....

THEN MKIGF = 13-11-9-7-6

Q 229. when numbers are written in base b, we have 12x25=333. The value of b is

a) 6 b) 8 c) 7 d) none of these

Solution- when a no. is written in base b

(1*b^1+2*b^0)*(2*b^1+5*b^0)=(3*b^2+3*b^1+3*b^0)

(b+2)*(2b+5)=(3b^2+3b+3)

solve it further a quadratic eq. is formed

b^2-6b-7

after solving it can be written as

(b-7)(b-1)

therefore b=7

b=7

Q 230. How many polynomial functions f of degree >=1 satisfy f(X2) = (f(X)) 2 =f (f(X)).

a) 1 b) 0 c) 2 d) more than 2

Solution- :Let f(x)=x^2

f(x^2)=x^4

[f(x)]^2=x^4

f(f(x))=x^4

Only 1 sol.

Q 231. A cow and a horse are bought for Rs.200000. The cow is sold at a profit of20% and the horse is sold at a loss of 10%. The overall gain is Rs. 4000.The cost pric of the cow is:

a) Rs 130000

b) Rs 80000

c) Rs 70000

d) Rs 120000

Solution- let cost of cow be x

let cost of horse be 200000-x

so, (120%)x-(90%)(200000-x)=204000

solve the above equation vil get x=80000

thus cost price of cow is 80000/-

Q 232. Raj drives slowly along the perimeter of a rectangular park at 24kmph and completes one full round in 4mins. The ratio of length is to breadth is 3:2. What are its dimensions?

a) 450m*300m

b) 150m*100m

c) 480m*320m

d) 100m*100m

Solution- The ratio of length is to breadth is 3:2

So,

Length is 3x and breadth is 2x

Now, The perimeter of the rectangular park is; 2(Length + Breadth)

2 (3x+ 2x)

2*5x

=10x

Raj drives slowly along the perimeter of a rectangular park that's mean he covered the total distance 10x.

So, the total distance is = 10x

Total time 4mins, so 4/60 hrs.

Speed 24kmph

We know that,

Distance = Time * Speed

Distance = 4/60 * 24

Distance= 1.6

Now, we can write;

10x= 1.6

x = 1.6/10

x = 0.16

Length is (3*0.16) = .48km or 480 m

Breath (2*0.16) = .32km or 320 m

Ans. Its dimensions are 480m * 320 m

Q 233. or which of the following n is the number 2^74 + 2^2058 + 2^2n is a perfect square.

a) 2010 b) 2018 c) 2012 d) 2020

Solution- 2^74 + 2^2058 + 2^2n =(2^37)^2 + (2^1029)^2 + (2^n)^2

now, if we put 2^37 = a ; 2^1029 = b; Then for the above expression to be perfect square 2^2n must be equal to (2*a*b)= 2*(2^37)*(2^1029);

==> 2^2n = 2^(1067)

==> 2n = 1067 ,

but this case is not possible since R.H.S is an odd integer whereas L.H.S is an even integer.

So , the above mentioned case can't hold.

Now,if we put 2^37 = a; 2^n = b ; So, for the given expression to be perfect square 2^2058 = (2*a*b)= 2*(2^37)*(2^n) = 2^(n+38);

So, 2058 = (n+38)

=> n = 2020

So,The answer is n = 2020

Q 234. Father is 5 times faster than son. Father completes work in 40days before son.If both of them work together when will work get complete?

a) 8 days b) 50/6 days c) 10 days d) 20 days

Solution- et takes x days TO COMPLETE THE WORK than father will take x-40

as we know time is inversely propotional to speed so

time=1/speed

fatherstime/sonstime

x- 0/x

and given that speed ration btwn father and son is 5:1

so x-40/x=5/1

x-40=5x

4x=40

x=10

so if father and son work together than

1/10+1/50

25/3 days is ans

Q 235. a beaker contains 180 litres of alcohol. On day 1, 60 litres of alcohol is replaced with water. On 2nd and 3rd days 60 litres of the mixture in the beaker is replaced with water. What will be the quantity of alcohol in the beaker after 3rd day?

a) 40 liters b) 80 liters c) 53.33 liters d) 100 liters

Solution-

1st time 180-60=120

2nd time 120-60*120/180=80lit

3rd time 80-60*80/180= 160/3 lit

So c) 53.33 liters

Q 236. A man asks 5 people to make a guess about the amount of money in his pocket which is less than 50.

A guess that the amt is a multiple of 10..

B guess that the amt is a multiple of 12..

C guess that the amt is a multiple of 15..

D guess that the amt is a multiple of 18...

E guess that the amt is a multiple of 30

Which of the following guesses are correct?

a)AE

b) AB

c) BC

d) DE

Solution- AE:30 is multiple of both 10 and 30 which is less than 50......**possible answer.**

AB: there is no multiple of 10 and 12 less than 50. .....**not possible answer.**

BC: Same logic as that of AB......**not possible answer.**

DE: Same logic as that of AB...... **not possible answer.**

Hence answer is AE.

Q 237. Letters of alphabets no from 1 to 26 are consecutively with 1 assigned to A and 26 to Z. By 27th letter we mean A, 28th B. In general 26m+n, m and n negative intezers is same as the letters numbered n.

Let P = 6, strange country military general sends this secret message according ot the following codification scheme. In codifying a sentence, the 1st time a letter occurs it is replaced by the pth letter from it. 2nd time if occurred it is replaced by P^2 letter from it. 3rd time it occurred it is replaced by p^3 letter from it. What is the code word for ABBATIAL

a) GHNNZOOR

b) GHKJZOHR

c) GHHGZOGR

d) GHLKZOIR

Solution-A should be coded as 1+6 = G (it occurred for first time)

B should be coded as 2+6 = H (it occurred for first time)

B Should be coded as 2 + 36 = 38 - 26 = 12 = L (it occurred for second time)

Option d)

Q 238. 30L + 3Q = 1167 ,30L + 6Q = 1284

Find L?

a) 30 b) 35 c) 40 d) 45

Solution- 30L+3Q=1167 ---- (i)

30L+6Q=1284 ---- (ii)

Solving (i) & (ii),Q=39, L=35

Q 239. The price of a book in four different shops and the successive discounts offered for the books is given below. Select the option in which the price of the book is the least.

(a) 10%, 5%, and 5% discount on Rs.195

(c) 12.5% and 12.5% discounts on Rs.205

(b)25%, discount on Rs.200

(d) 10%, and 15% discounts on a marked price of Rs.190

Solution- we know that there is a formula on discount of m% and n% respectively which is(m+n-mn/100)%.According to this for first case,total discount is 18.775%,so after discount the price of the book is Rs.158.38.For second case total discount is 23.4375%,so the price of the book after discount is Rs.156.95.For third case after discount the pice of the book is Rs.150.For last case total discount is 23.5% and the price of the book after discount is Rs.145.35.

So ans is option d in which the price is least.

Q 240. Divide 50 into two parts such that the sum of their reciprocals is 1/12. the numbers are

a) (24,26) b) (28,22) c) (27,23) d) (20,30)

let one number be x

the other number will be 50-x

1/x + 1/(50-x)= 1/12

50-x+x = x(50-x)/12

600 = 50x-x^2

x^2-50x+600=0

x^2-30x-20x+600=0

x(x-30)-20(x-30)=0

(x-30)(x-20)=0

x= 20or 30 the numbers are 20 and 30

Q 241. John told mark that if mark gives 1/3 rd of his money to him, he will have rs. 75. Mark told john that if john gives half of hi money to him, he will have rs. 75. How much money did they totally have?

a) 105 b) 125 c) 150 d) 75

Solution- let mark be y nd john be x

x+y/3=75

y+x/2=75

by solving

x=60

y=45

total=105

Q 242. A city in the U S has a basketball league with three basketball teams, the Arêtes, the Braves and the Deities. A sports writer notices that the tallest player of the Arêtes is Shortest than the shortest player of the Braves. The shortest of the Deities is shortest of the Arêtes, while the tallest of the braves is taller than the tallest of the Arêtes. Which of the following can be judged with certainty?

X) Paul, a brave is taller than David, an Arêtes.

Y) David, a Deities, is shorter than Edward, an Arêtes.

a) X only

b) Both X and Y

c) Neither X and Y

d) Y only

Solution- its clearly written that the tallest player of the ArÃªtes is Shortest than the shortest player of the Braves.

=> all players of braves are taller than all players of Arates.

=> X is true

=> ans = (a)

Q 243. Sum of the digits in the equation (1^100)*(125^135) is

a) 2 b) 5 c) 3 d) 8

Solution- 16^100 * 125^135 => (2^4)^100 * (5^3)^135 => 2^400 * 5^405 = 2^400 * 5^400 * 5^5

= (2*5)^400 *3125

=10^400 * 3125

sum of 10^400= 1+0 + 0+ 0+..... = 1

sum of 3125 = 11

therefore= 1 * 11 = 11

therefore = 1 + 1= 2

ans is 2

Q 244. what is the average of the first 200 terms of the series 1,-2,3,-4,5,-6,7....

a) -0.5 b) -50.5 c) 0.5 d) 50

Solution- average of the first 200 terms of the series 1,-2,3,-4,5,-6,7...

=1+(-2)+3+(-4)+...+199+(-200)/200

=(1-2)+(3-4)+(5-6)+...+(199-200)/200

=(-1)+(-1)+(-1)+. 100times/200

=-100/200

=-0.5

Q 245. A sequence x1,x2,x3…. Is said to be in harmonic progression if the reciprocals 1/x1,1/x2,1/x3… are in arithmetic progression. The 5th and 7th term for harmonic progression are 30 and 50 respectively what is difference between 6th and 4th term ?

a) 16 b) 14.5 c) 13.4 d) 12.5

Solution- hp series is

a,a/1+d,a/1+2d,.....

so applying this

5th term is 30=a/1+4d

and 7th term is 50=a/1+6d

so from this two eq. we get

a=50/3

and d=-1/9

putting this two for 4th and 6th term we get 4th term =25 ..6th term=37.5

difference=12.5

Q 246. Each of A,B and C need a

certain unique time to do certain work. C needs 1 hour less than A

to complete the work. Working together they require 30 minutes to

complete 50% of the work. The work also gets completed if A and B

start working together and A leaves after 1 hour and B works further

3 hours. How much work does C do per hour?

a) 16.66%

b) 66.66%

c) 50%

d) 33.33%

Solution- 50% suppose A do work in x hrs ,B in y, then c would do in x-1 hrs..then 1/x+1/x-1+1/y=1 in 1 hrs..also A&B 1 hrs work 1/x+1/y then work remaining xy-x-y/xy which is done by B in 3 hrs so (xy-x-y)/xy=3/y it results y=4x/x-1 putting value we get x=3 so y=6 so A CAN DO WORK in 3 hrs B in 6 hrs & C IN 2 hrs that is 50%

Q 247. In an year N, the 320th day of the year is Thursday. In the year N+1 the 206th day of the year is also Thursday. What is the 168th day of In the year N-1?

a) Friday

b) Thursday

c) Tuesday

d) Saturday

Solution- riday..If the Nth year is a non-leap year, then 320th day of year N to 206th day of year N+1 is [(365

– 320) + 206] = 251 days i.e. 35 weeks + 6 days. But this will not make the two days to be

Thursday.

Thus, the Nth year has to be a leap year. In this case 320th day of year N to 206th day of year

N+1 is [(366 – 320) + 206) = 252 days i.e. 36 weeks. Hence both days will be same day of the

week i.e. Thursday as given by the data.

168th day of N-1 year to 320th day of N year is [(365 – 168) + 320) = 517 days i.e. 73 weeks +

6 days. Thus, if the 320th day is Thursday, then the 168th day of year will be Friday, option

Q 248. What is the remainder when 6^17+17^6 is divided by 7?

a) 1

b) 6

c) 0

d) 3

Solution- 6^17+17^6 = (7-1)^17 + (21-4)^6 = (7-1)^17 + (7*3 -4)^6.…[eqn1]

if the [eqn1] is expanded then every term of the expansion except [(-1)^17 + (-4)^6] will have 7 as one of its factors.

Just think a little bit about the binomial expansion of both [(7-1)^17] and [(7*3 -4)^6] ,then u can readily point out that only the last term of both the expansions , i.e , [(-1)^17] & [(-4)^6] respectively,will don't have 7 as one of its factors.

So, we have to calculate the remainder when [(-1)^17 + (-4)^6] is divided by 7.

Now, clearly (-1)^17 = -1

and, (-4)^6 = 4^6 = 2^12 = (2^3)^4 = (7+1)^4.

Now, a same reasoning related to binomial expansion mentioned previously explains why, when (7+1)^4 is divided by 7 will leave a remainder 1.

So, (7+1)^4 will be of the for (7*A + 1); where A is some +ve integer, to know whose value isn't important in this case.

So, when

[(-1)^17 + (-4)^6] will be divided by 7

or, [-1 + (7+1)^4] will be divided by 7

or, when [-1 + 7*A +1] will be divided by 7

or, when [7*A] will be divided by 7 , clearly therefore the remainder will be zero, i.e, 0.

So, the answer is OPTION 3)0.

Q 249. George and Mark can paint 720 boxes in 20 days. Mark and Harry in 24 days and Harry and George in 15 days. George works for 4 days, Mark for 8 days and Harry for 8 days. The total number of boxes painted by them is

a) 252 b) 516 c) 348 d) 492

Solution- Capacity of G + M = 720 / 20 = 36

M + H = 720 / 24 = 30

H + G = 720 / 15 = 48

Combined capacity = 2 (G + H + M) = 114

G + H + M = 114 / 2 = 57

Now capacity of G = (G+H+M) - (H + M) = 57 - 30 = 27

M = (G+H+M) - (H + G) = 57 - 48 = 9

H = (G+H+M) - (G + M) = 57 - 36 = 21

Given that G worked for 4 days, and mark for 8 and harry for 8 days

So total work by them = 4 x 27 + 8 x 9 + 8 x 21 = 348

Q 250. Megha drives along the perimeter of square field of side 10kms.She drives along the first side at 10kmph.along second side 20 kmph along 3rd side 30kmph and along the forth side at 40 kmph.Her avearse speed is ?

a) 19.2kmph b) 18kmp c) 30kmph d) 20kmph

Solution- Average speed : total distance/total time

Total distance=10*4=40 km

total time =10/40 + 10/20 + 10/30 + 10/40

So, Average Speed = 19.2 Km/h

Q 251. For real number x, int(x) denotes integer part of x. int(x) is the largest integer less than or equal to x.int(1,2)=1, int(-2,4)= -3. Find the value of int(1/2)+int(1/2+100)+int(1/2+2/100)+.....+int(1/2+99/100)

a) 50 b) 49 c) 51 d) 48

Soluiton- if it is int(1/2 + 100) the answer is 150 lest if it is int(1/2 + 1/100) the answer is 50.

from int(1/2 + 49/100)= 0 and from int(1/2 + 50/100)= 1 and int(1/2 + 99/100) = 1 so from 50 to 99 it is equal to 50

Q 252. 10 years ago 10 people age was33. After 3 years a person of age 40 dies. After another 3 years anothe person of 40years dies.After another 3 years another person of 27years dies. Find the present average age?

a) 43 b) 44 c) 35 d) 40

Solution- 10 year ago 10 people= 33

10 year ago total age=330

after 3 year 1 person with age 40 died = ie take his age as 37 before 3 years

lly for next 2 persons ; consider as 34 (40-6) and as 18(27-9) in 6 and 9 years ago(ie 37+34+18=89) 10 years ago age of 7 people =330-89=241

nw consider present age 7*10=70+241=311

nw avg= 311/7=44.43(ans)

Q 253. Raj invested in Indigo, HUL and SBI shares at Rs. 300,

Rs. 200 and Rs. 5 per share, 100 shares for Rs. 10000. The number of Indigo and HUL shares he bought are

a) 15, 25 b) 23, 17

c) 17, 23 d) 17, 60

Solution- ans will be 17,23 coz..

left sbi share will be 100-(17+23)=60

then indigo=300*17=5100

and HUL=200*23=4600

sbi=5*60=300

.....

total=5100+4600+300=10000

Q 254. Raju can do a piece of work in 10 days..vicky 12days,tinku 15 days..day all start the work together,but raju leaves aftr 2 days,vicky leaves 3 days before the work is completed..how many days work is completed?

a) 7 b) 5 c) 9 d) 6

Solution- raju+vicky+tinku one day work = (1/10)+(1/12)+(1/15)= 1/4

now for 2 days together work done= 2*(1/4)= 1/2

.....now work done by tinku in last 3 days alone=(3/15)=(1/5)

remaining work= 1-(1/2 + 1/5)= 3/10

....now vicky+tinku one day work=(1/12)+(1/15)=9/60

therefore 3/10 work will be done by both of them in= (60/9)*(3/10)= 2 days

now answer= 2days(as got from above; it is days required to do rem. work)+ 3days(when vicky leaves)+2days(raju leaves)=7 days ans.

Q 255. Box of fruits can be loaded into a truck in 9 minutes by a

worker. And 8 boxes will fill a truck completely. How many trucks

can be loaded in 1 ½ hours if there are 16 men together.

a) 1

b) 2

c) 3

d) 0

Solution- 1 worker can in 9 mins load 1 box

1 worker can in 90 mins load (1*90/9)=10boxes

16 worker can in 90 mins load (10*16)=160 boxes

So no. of trucks needed=160/8=20 i think...

Q 256. the first 44 positive integers are written in order to form the largest number N=12345678910111213..424344. what will be the he remainder when N is diveded by 45?

a) 4 b) 9 c) 14 d) 18

Solution- no will be div by 45 when it is divided y both 9 and 5

a no is div by 9 if sum of digits is div by 9

now sum of digits of given no is 270

so the no is div by 9

and since last digit is 4 when div by 5 it leaves rem 4

let us say numbers such as 9,54...

we see they are div by 9 but leaves rem 4 when div by 5..

so when div by 45 they leave rem 9

so req ans=9

Q 257. The sum of 3 consecutive numbers of the four numbers A, B, C, D are 4613,4961,5010,5099 then what is the largest number among A,B,C,D ?

a) 1948 b) 1463 c) 1601 d) 1550

Solution- Let, S1,S2,S3,S4 be the sums of A,B,C,D taking 3 of them at a time.

So, by the given data in the problem,it follows that :

S1+S2+S3+S4 = 4613+4961+5010+5099

=>S1+S2+S3+S4 = 19683....[eqn 1]

Now, Without the loss of generality we can assume

S1= A+B+C

S2= B+C+D

S3= C+D+A

S4= D+A+B

----

adding the above terms we get,

S1+S2+S3+S4 = 3(A+B+C+D)

=>19683 = 3(A+B+C+D); [From [eqn 1]]

=>(A+B+C+D)=(19683/3)

=>(A+B+C+D)=6561

So,the greatest term among them =sum of all four - sum of lowest three terms

So,the greatest term among them=6561-4613 = 1948.

So,the greatest term among them = 1948

So, the correct answer is : OPTION a)1948

Q 258. Three cars A, B and C are participating in a race.A is twice as likely as B to win and B is thrice as likely as C to win. What is the probability that B will win, if only one of them can win the race?

a) ½ b) 2/5 c) 3/10 d) 1/10

Solution- p(A)=2*p(B)

P(B)=3*p(c)

P(A) +P(B)+P(C) =1

2 * P(B) + P(B)+P(B)/3 =1

10*p(B) /3=1

P(B)=3/10

Q 259. 3 white chips, 7 blue chips, 16 green chips, 2 chips drawn from the box in succession what is the probability that one is blue and other is white?

a) 7/50 b) 8/30 c) 7/25 d) 21/25*13

Solution- no of ways that 2 chips selected from total of 26 is 26C2=26*25/2=13*25

no of ways that 1 white and 1 blue is selected is 3C1*7C1=21

total probability=21/13*25

Q 260. George is 2/3 rd as efficient as smith and smith is ¾ th as efficient as John. George working alone is what fraction of All of them working together.

a) 2/3 b) 2/9 c) 4/9 d) 1/3

Solution- g:s=2/3:1=2:3

s:j=3/4:1=3:4

g:s:j=2:3:4

so ans is 2/9

Q 261. A man can load one box in 9 minutes. A truck can contain 8 boxes. If 16 men load for one and a half hour, how many trucks will be loaded?

a) 20 b) 10 c) 15 d) 40

Solution- 1 person in 9 min can load 1 box

1 person in 1 min can load 1/9 boxes

16 person in 1 min can load 16/9 boxes

16 person in 90 min can load 16*90/9=160 boxes

now

8 boxes in 1 truck

so

160 boxes comes under 20 trucks

so answer is 20 trucks loaded

Q 262. In the simple subtraction problem Below , sum single digits (not necessarily distinct) are replace by letters , find the value of 7*A + 7*B +6*C*D

A 7 C 2

- 4 B 6 8

_____

5 4 3 D

a) 77 b) 95 c) 84 d) 70

Solution- A7C2

-4B68

_____

543D

So after solving the equation we get A=9,B=2,C=0,D=4(because no are not distinct so it may be repeated).

so Ans of 7A+7B+6CD=7*7+7*2+6*0*4=77

Q 263. There are 5 distinct integers a,b,c,d,e in ascending order.

(68-a)(68-b)(68-c)(68-d)(68-e)=725.

then what is the value of a+b+c+d?

a) 34 b) 136 c) 306 d) 238

Solution- 725 = -5*-1*1*5*29

so we can write,

68-e = -5

=> e = 73

68-d = -1

=> d = 69

68-c = 1

=> c = 67

68 - b = 5

=> b = 63

68-a= 29

=> a = 39

so a+b+c+d = 39+63+67+69 = 238

Q 264. A circle has 29 points arranged in a clockwise manner numbered from 0 to 28, as shown in the figure below. A bug moves clockwise around the circle according to the following rule. If it is at a point i on the circle, it moves clockwise in 1 second by (1 + r) places, where r is the reminder (possibly 0) when i is divided by 11. Thus if it is at position 5, it moves clockwise in one second by (1 + 5) places to point 11. Similarly if it is at position 28 it moves (1 + 6) or 7 places to point 6 in one second. If it starts at point 23, at what point will it be after 2012 seconds?

a) 1 b) 7 d) 15 d) 20

Solution- after 1st second : (1+23%11 = 1) = 2 places [25]

after 2nd second : (1+25%11 = 3) = 4 places [0]

after 3rd second : (1+0%11 = 0) = 1 place [1]

after 4th second : (1+1%11 = 1) = 2 places [3]

after 5th second : (1+3%11 = 3) = 4 places [7]

after 6th second : (1+7%11 = 7) = 8 places [15]

after 7th second : (1+15%11 = 4) = 5 places [20]

after 8th second : (1+20%11 = 9) = 10 places [1]

now,for the same pattern from 4th sec to 8th sec will repeat itself (5 sec intervals)..

total time = 2012 secs

first 3 secs out of pattern...so time left 2012 - 3 =2009 secs

now no. of repetitions in the leftover time = 2009/5 = 401. .. remainder = 4

for the next 4 iterations following the similar pattern

the position will be 20..

Q 265. Total income of Eesha in the years 2003,2004,2005 was $36,400.Her income increased by 20% each year.What was her income in 2005?

a) 14500 b) 14000 c) 14200 d) 14100

Solution- let in 2003 income=x.so in 2004 it will be (x+x*20%)=6x/5.and in 2005 it will be(6x/5+6x/5*20%)=36x/25.

x+6x/5+36x/25=36400 then x=10000.in 2004 income=12000 and in 2005 income=14400

Q 266. Length, Breadth and Height of a cuboid is in the ratio 1:3:27. Volume of the cuboid is 27 m3. If the length is doubled and Breadth & Height are halved, then what is the change in the volume of the cuboid?

a) Decreased by 15% b) Decreased by 18% c) Decreased by 30%

d) Decreased by 50%

Solution- given l:b:h=1:3:27

so now volume will be lbh=1*3*27=81

now given l is doubled so l= 2l

b and h are halved so b=3b/2,h=27h/2

now new volume 2l*b/2*h/2

so substitute values then u get lbh=2*(3/2)*(27/2)=40.5

so the change is 81-40.5=40.5

so half of the original volume

Q 267. Two people, Ranbir and Katrina decide to meet at a a beach between 1 pm to 2 pm, given that both will surely turn up once in the given time frame. If Ranbir arrives, he waits for 15 minutes and then leaves feeling betrayed and similarly Katrina waits for 15 minutes after she arrives. So what's the probability that they meet?

a) 0.35 b) 0.45 c) 0.25 d) 0.15

Solution- waiting time of both katrina and runbir is 15 min. Means they can meet only meet each other in 15 Min. , if they 1 more min. than they can't meet each other .. so chances of meeting in only(Favourable event)=15 min

Total no. of min.(Sample space)= 60 min

Probability of Meeting is =(Favourable event)/(Sample space)

15/60=0.25 ans.

Q 268. If (3a+6b)/(5a+12b)=12/23 determine the value of 3a2+5b2/ab

a) 19/2 b) 32/3 c) 9 d) 31/3

Solution- consider (3a+6b)/(5a+12b)=12/23 --- >eq 1

sol eq-- >1 the v vil get a/b=2/3

div 'ab 'to the num and den for 3a2+5b2/ab

therfore, 3*(a/b)+5*(ba)

sub a/b=2/3 in ab eqtn

v get 19/2

Q 269. A series of books was published at 10years intervals when the 10th book was issued the sum of publication years was 19,560when was the 1st book published

a)1910 b)1914 c)1911 d)1909

Solution- 19560=10/2[a+a+(10-1)*10]

19560=5[2a+90]

19560-450=10a

a=19110/10

a=1911

Q 270. given the digits 1,3,6,9 find the probability that a 3 digit number formed by using them with no digit repeated is divisible by 4

a) 1/12 b) 1/4 c) 1/6 d) none

Solution- divisibility rule for four is last 2 digit must be divisible by 4. so last 2 digit must be 16,36,96.

(1) -- -- 1 6,(2) -- -- 3 6 , (3) ----- 9 6

mow you can put in 1 st case 9,3 yoy have 2*1 choice, similar in 2nd and 3rd case ..so total number of choice is =6

and total no. odf case = 4!=24

p(divisible 4)= 6/24=1/4

Q 271. The diagonal of a square is twice the side of equilateral triangle the ratio of Area of the Triangle to the Area of Square is?

a) √3:8 b) √2:5 c) √3:6 d) √2:4

Solution- diagonal of a square is 2a

area of the square=1/2*(diagonal)^2 i.e 2a^2

side of the equi triangle is a

area of equi triangle=(3/4)a^2

3/4a^2:2a^2

3:8

Q 272. The number of different non congruent triangles with integer side and perimeter 15 is

a) 9 b) 7 c) 10 d) 6

Solution- B)7, Let ,1st side of triangle =a, 2nd side =b & 3rd side =c

A/C to law of triangle a+b>c so (a+b) must be 8 or 9 or 10 or 11 or 12or 13 or 14 but not be 15,

Q 273. The number of committes of size 10 that could be formed from 10 men & 10 women such that committee has atleast 6 women is

a)60626 b) 210 c) 10210 d) none

Solution- Option: (a)

10C6x10C4 + 10C7X10C3 + 10C8X10C2 + 10C9X10C1 + 10C10X10C0

= 44100+14400+2025+100+1

= 60626

Q 274. How many positive multiples of 10 that are less than 1000 are the sum of 4 consecutive integers

a) 51 b) 50 c) 49 d) none

Solution- in the question what they given is we have to find the sum of the 4 consecutive numbers that sum should be multiple of like 10 ie..10,20. ... and should be less than 1000.

so that we can conclude tat last 10 multiple before 1000 is 990

then w.k.t 1+2+3+4=10

6+7+8+9=30--->2

11+12+13+14=50

16+17+18+19=70

like wise it will be repeat,

based upon the given we need to find upto 990 how many time it occurs is

so that use this formula to find last number in AP

sn=a+(n-1)d

we knw sn=990=>last number

a=10=>initial number

d=20==>diff b/w two consecutive numbers

990=10+(n-1)20

990=10+20n-20

990=20n-10

20n=1000

n=50

Q 275. Inabhi's class of 44 students,28 students speak malayalam,26 students speak tamil,9 students speak none of the two languages.How many students speak both tamil& Malayalam

a) 54 b) 38 c) 19 d) 10

Solution- (44-9)=28+26-n(A$B)

35=54-n(A$B)

n(A$B)=19

Q 276. The sum of two numbers is 2016 and their product is 32 the sum of their reciprocals is

a) 63 b) 9 c) 32+ 2014 d) 32- 2014

Solution- let two no are x,y

x+y=2016..... **(1)**

xy=32 .....**(2)**

therefore

1/x+1/y=x+y/xy=2016/32=63

Q 277. The sum of two numbers is 45. The sum of their quotient and is reciprocal is 2.05, the product of the numbers is?

a) 450 b) 205 c) 400 d) 500

Solution- a + b = 45

a/b+b/a = 2.05

=>(a^2+b^2)/ab = 2.05

=>((a+b)^2−2ab)/ab=2.05

=>(a+b)^2 = 2.05ab + 2ab = 4.05ab

=> ab = 45^2 / 4.05 = 500

Q 278. Find the number of divisors of 1728.?

a) 28 b) 20 c) 30 d) 18

Solution- 1728= 2^6 * 3^3

Hence the Number of factors = (6+1) x (3+1) = 7 x 4 = 28.

We know that if a number represented in standard form (a^m *b^n) , then the number of factors Is given by (m+1)(n+1).

Answer is 28

Q 279. In 4 years, Rajs father will be double Rajs age then. Two years ago, while his mother was twice his age that time. If Raj is going to be 32 years old 8 years from now, then what is the sum of his parents age now?

a) 96 b) 100 c) 102 d) 98

Solution- R.F=RAJS FATHER AGE

R=RAJS AGE

R.M=RAJS MOTHER AGE

(R.F+4)=2*(R+4)

(R.M-2)=2*(R-2)

R+8=32

therefore, R=24

therefore, R.M=46

therefore, R.F=52

R.F + R.M = 98

Q 280. when a no. divided by 406 it gives 115 as a remainder. when it divided by 29 what is the remainder.???

a) 27 b) 7 c) 28 d) 19

Solution- let x is divided by 406 & gives quotient y & remainder 115 then

x = 406 * y + 115

now, x / 29 = (406 * y + 115)/29 = 406 * y/29 + 115/29

=> rem = 115/29 = 28

Q 281. Raj divided 50 into two parts such that the sum of their reciprocal is 1/12, we get the parts as

a) 28, 22

b) 24, 36

c) 36, 14

d) 20, 30

Solution- (1/20)+(1/30)=5/60=1/12 so ans is (d) 20,30

Q 282. p/q - q/p =21/10

then

4p/q + 4q/p = ?

a) 58/10 b) 113/10 c) 58/5 d) 121/110

Solution- 58/5

let us take p/q=x and q/p=1/x

x-1/x=21/10

then by solving we get a quadratic equation like 10x^2-10-21x=0

10x^2-25x+4x-10=0

then by taking factors we get two values for x,x=-2/5 and x=5/2

by substituting x=5/2

4(5/2)+4(2/5)=58/5

Q 283. what is the 2015th no. of problemsolvingproblemsolving.....

a) g b) p c) n d) o

Solution- "problemsolving"it is a 14 letters word

total no=2015th

"g" is repeat=2015/14=143 remainder=13

13th words is "N"

Q 284. In the sixth, seventh, eighth, and ninth basketball games of the season, a player scored 23,14, 11, and 20 points, respectively. Her points-per-game average was higher after nine games than it was after the first five games. If her average after ten games was greater than 18, what is the least number of points she could have scored in the tenth game?

a) 29 b) 30 c) 28 d) 27

Solution- The sum of the scores for games 6 through 9 is 68. The average in these four games is \frac{68}{4} = 17

The total points in all ten games is greater than $10\cdot 18 = 180. Thus, it must be at least 181.

There are at least 181 - 68 = 113 points in the other six games: games 1-5and game 10.Games 1-5 must have an average of less than 17. Thus we cannot put more than $16 + 17 + 17 + 17 + 17 = 84 points in those five games.

Thus, the tenth game must have at least $113 - 84 = 29 points

Q 285. 2/3rd of the balls in a bag are blue, the rest are pink. if 5/9th of the blue balls and 7/8th of the pink balls are defective, find the total number of balls in the bag given that the number of non defective balls is 146.

a) 216 b) 649 c) 432 d) 578

Solution- let the total balls be x

2/3x=blue & 1/3x=pink

non defective balls=146

total no. of balls=defective+non defective

hence defective=total-non defective(146)

(2/3x)*5/9+(1/3x)*7/8=t-146

0.66203x=t-146

146=0.3303x

x=432